

Overview of Sensors For Robotics

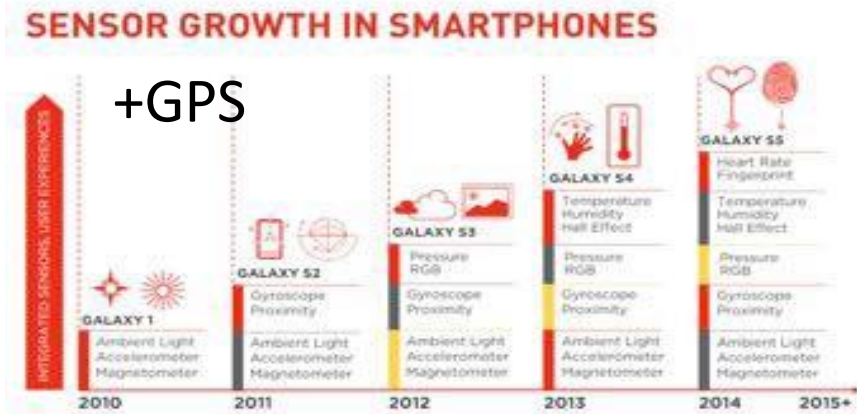
many slides from Siegwart, Nourbaksh, and
Scaramuzza, Jana Kosecka, Antonio Masegoza

Cristina P Santos, 11/10/2021

Robot Sensors

- Why do Robots Need Sensors?
- What can be Sensed?
- What Sensors are Out There?
- What can They do?
- How Much do They Cost?
- How Easy are They to Use?

Sample of everyday sensors



www.qualcomm.com



3D



PIR

Passive Infrared sensor

Camera



US

Radar

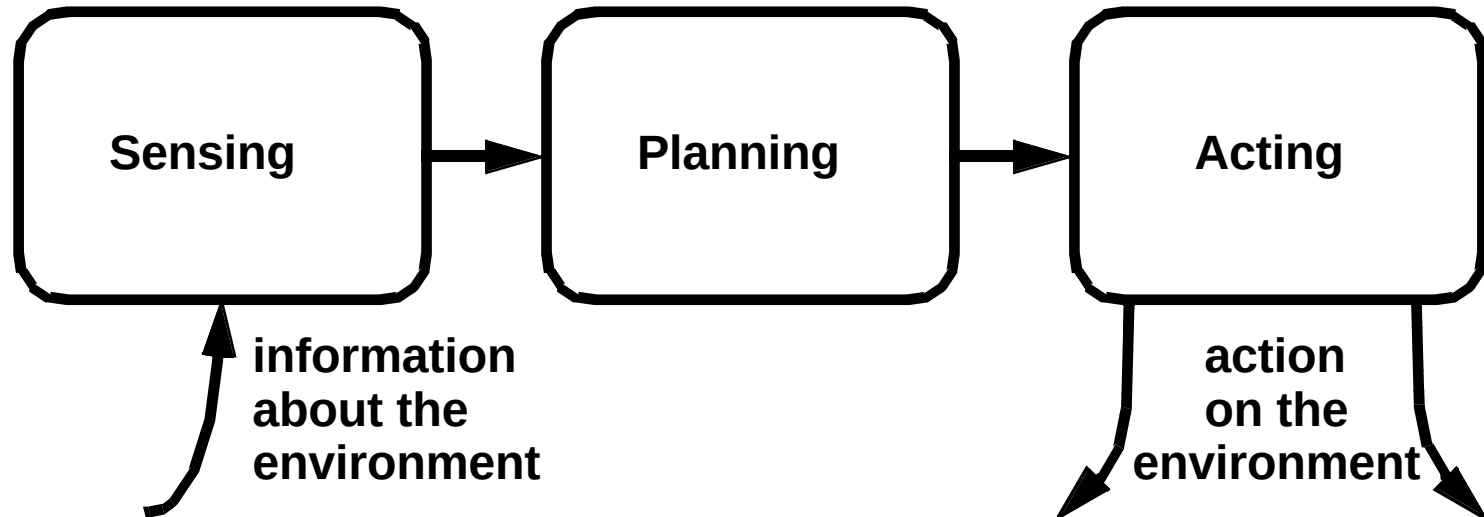


So far: Robotics always benefits, never drives sensor development

Sensors for Mobile Robots

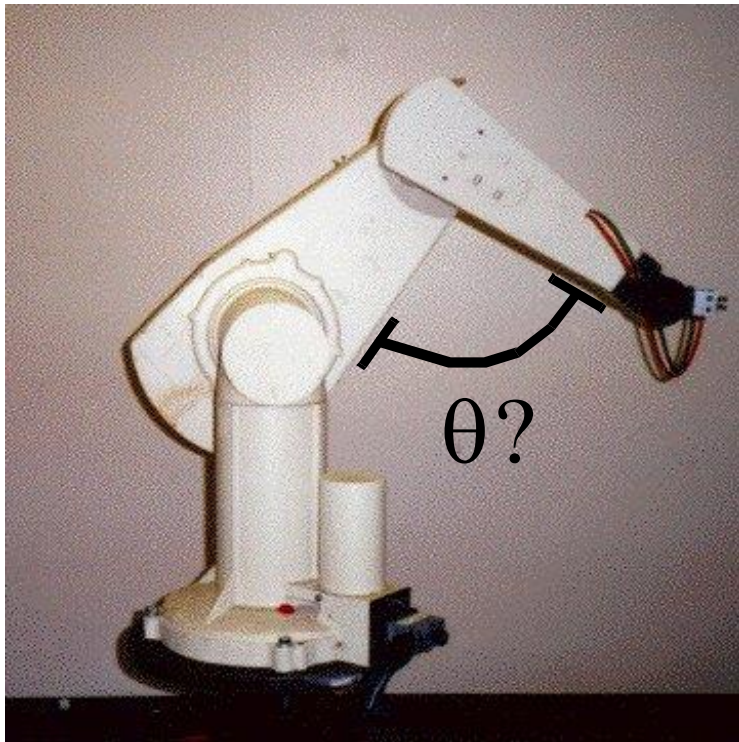
- Why should a robotics engineer know about sensors?
 - Is the **key technology** for perceiving the environment
 - **Understanding the physical principle** enables appropriate use
- Understanding the physical principle behind sensors enables us:
 - To **properly select** the sensors for a given application
 - To **properly model** the sensor system, e.g. resolution, bandwidth, **uncertainties**

What makes a machine a robot?



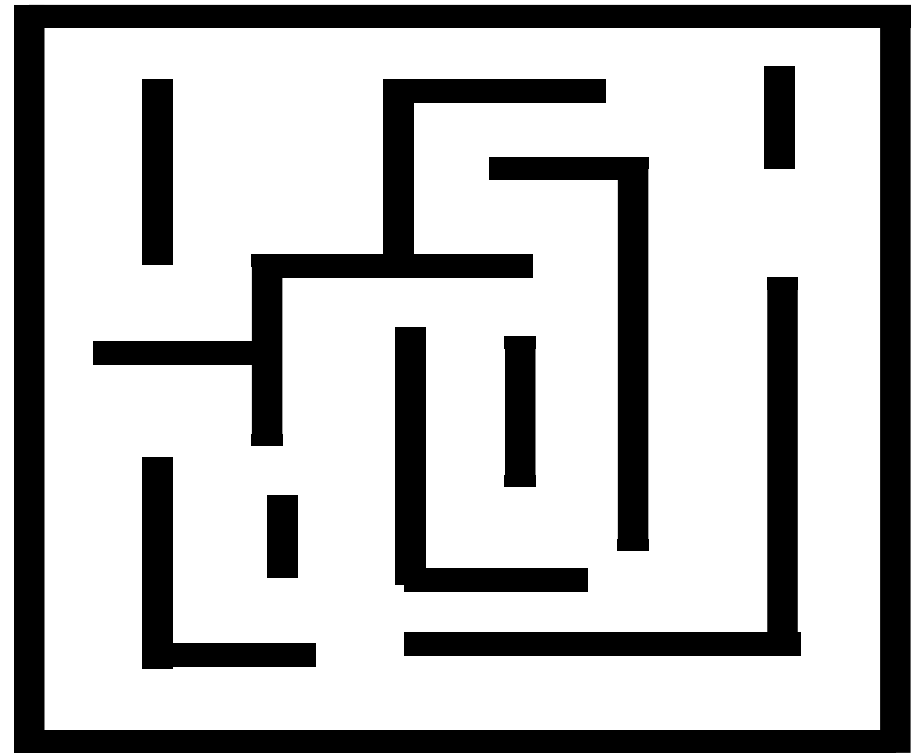
Why do robots need sensors?

**What is the angle
of my arm?**



internal information

Where am I?



localization

Sensing for specific tasks

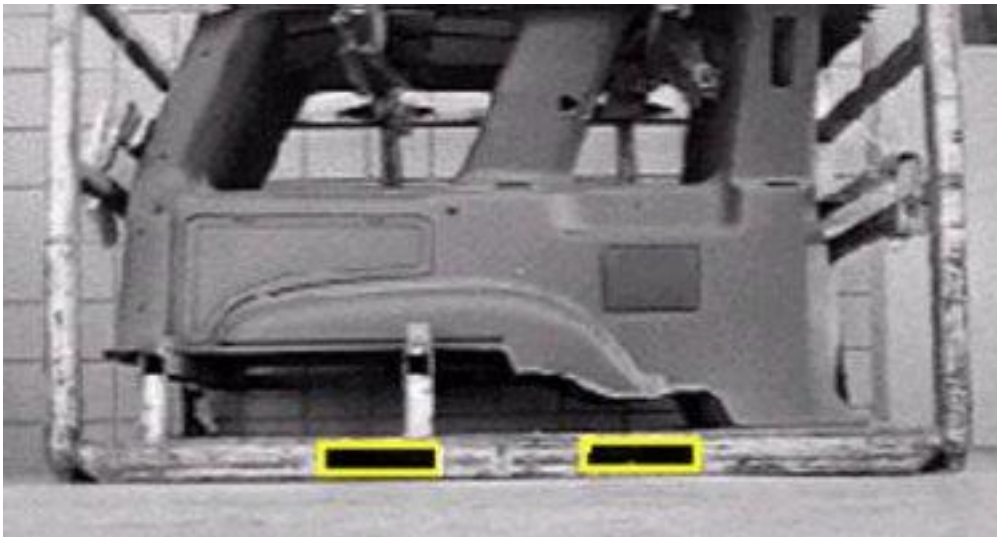
Where is the cropline?



**Autonomous
harvesting**

Sensing for specific tasks

Where are the forkholes?



Autonomous material handling

Sensing for specific tasks

Where is the face?



Face detection & tracking

Why Do Robots Need Sensors?

- Provides “awareness” of surroundings
 - What’s ahead, around, “out there”?
- Allows interaction with environment
 - Robot lawn mower can “see” cut grass
- Protection & Self-Preservation
 - Safety, Damage Prevention, Stairwell sensor
- Gives the robot capability to goal-seeking
 - Find colorful objects, seek goals
- Makes robots “interesting”

What is Sensing ?

- Collect information about the world
- Sensor - an electrical/mechanical/chemical device that **maps an environmental attribute** to a quantitative measurement
 - attribute mixtures - often no one to one map
 - hidden state in environment
- Each sensor is based on a ***transduction principle*** - conversion of energy from one form to another
- Also known as **transducers**

Different sensors

- differ **in their precision** and the **kind of data** that they provide, but none of them is able to completely solve the localization problem on its own.
- E.g. encoder measures position, but better on robotic arms.
for nonholonomic robot: motions that return the encoder values to their initial position, do not necessarily drive the robot back to its starting point.
- Different data: e.g.
 - accelerometer*** samples real-valued quantities that are digitized with some precision
 - odometer*** delivers discrete values that correspond to encoder increments.
 - vision sensor*** delivers an array of digitized real values (colors)

Classifying sensors

- Type of information
- Physical Principle
- Absolute vs. derivative
- Amount of information (Bandwidth)
- Low and high reading (Dynamic range)
- Accuracy and Precision

Functional Classification of Sensors

- What:
 - Proprioceptive sensors (*deal with self, internal*)
 - measure values internally to the system (robot),
 - e.g. motor speed, wheel load, heading of the robot, battery status
 - “where are my hands ?” (encoders, stretch receptors)
 - “am I balanced ?” (gyroscopes, INS)
 - Exteroceptive sensors (*deal with external*)
 - information from the robots environment
 - distances to objects, intensity of the ambient light, unique features.
 - “where is something ?”
 - “how does it look ?” (camera, laser rangefinder)
 - Interoceptive
 - what is my thirst level ? (biochemical)
 - what is my battery charge ? (voltmeter)

Functional Classification of Sensors

- **How:**
 - Passive sensors:
 - Measure energy coming from the environment
 - e.g. video cameras
 - Active sensors:
 - emit their proper energy and measure interaction of signal w/ environment
 - e.g. radar, sonar
 - better performance, but some influence on environment

Types of Sensors

- Classification by medium used
 - based on electromagnetic radiation of various wavelengths
 - vibrations in a medium
 - concentration of chemicals in environment
 - by physical contact

Sensors - What Can Be Sensed?

- Light
 - Presence, color, intensity, content (mod), direction
- Sound
 - Presence, frequency, intensity, content (mod), direction
- Heat
 - Temperature, wavelength, magnitude, direction
- Chemicals
 - Presence, concentration, identity, etc.
- Object Proximity
 - Presence/absence, distance, bearing, color, etc.
- Physical orientation/attitude/position
 - Magnitude, pitch, roll, yaw, coordinates, etc.

Sensors - What Can Be Sensed?

- Magnetic & Electric Fields
 - Presence, magnitude, orientation, content (mod)
- Resistance (electrical, indirectly via V/I)
 - Presence, magnitude, etc.
- Capacitance (via excitation/oscillation)
 - Presence, magnitude, etc.
- Inductance (via excitation/oscillation)
 - Presence, magnitude, etc.
- Other Things?

What Sensors Are Out There?

- Feelers (Whiskers, Bumpers) – Mechanical
- Photoelectric (Visible) – Active & Passive
- Infrared (light) – Active & Passive
- Ultrasonic (sound) – Active & Passive
- Sonic – Active & Passive
- Resistive/Capacitive/Inductive – Active & Passive

What Sensors Are Out There?

- Visual – Cameras & Arrays (Active & Passive)
- Color Sensors (Active & Passive)
- Magnetic (Active & Passive)
- Orientation (Pitch & Roll)
- GPS (location, altitude)
- Compass (orientation, bearing)
- Voltage – Electric Field Sensors
- Current – Magnetic Field Sensors
- Chemical – Smoke Detectors, Gas Sensors

Important Questions

- What are the sources of error?
- What are we measuring?
vs.
- What do we actually want to know?

Simple Practical Sensors that we can purchase

☐ Touch sensors

☐ Tilt sensors

☐ Encoders

☐ Bend sensors

☐ Light sensors

☐ Temperature sensors

☐ Potentiometers

☐ Laser rangefinders

☐ Cameras

A Closer Look

Touch Sensors - Feelers

Mechanical contact leads to ...

- closing/opening of a switch
- change in resistance of some element
- change in capacitance of some element
- change in spring tension ...

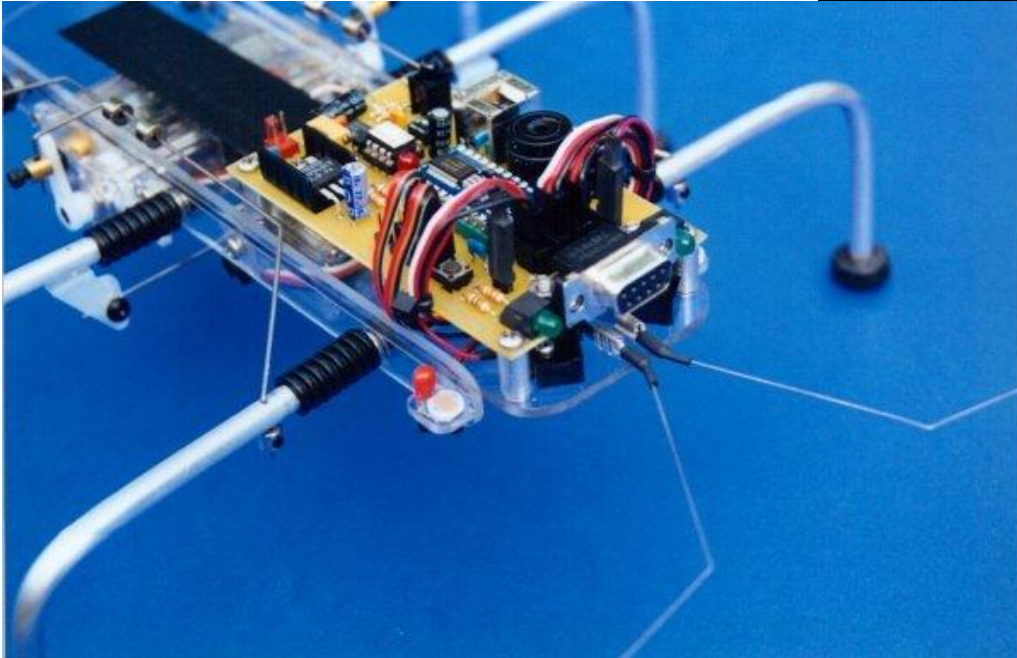
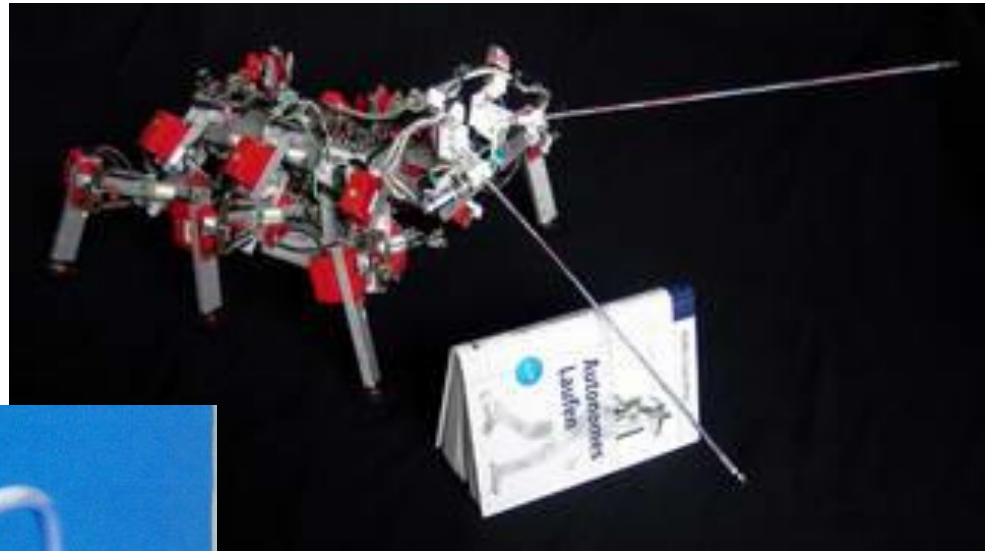
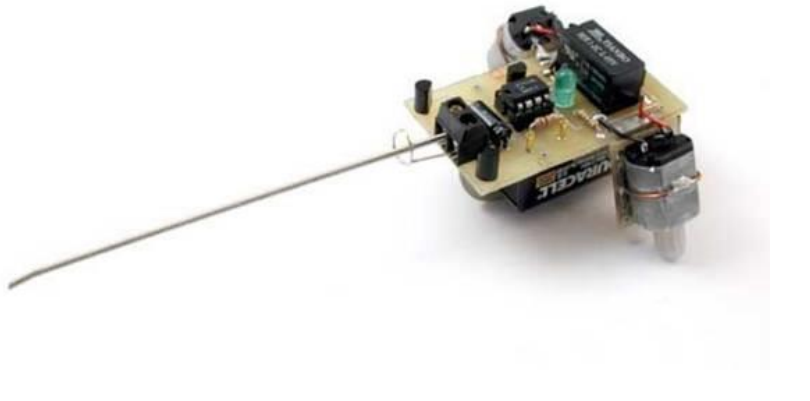
- Whiskers

- Piano wire suspended through conductive “hoop”
- Deflection causes contact with “hoop”
- Reaches beyond robot a few inches
- Simple, cheap, binary output

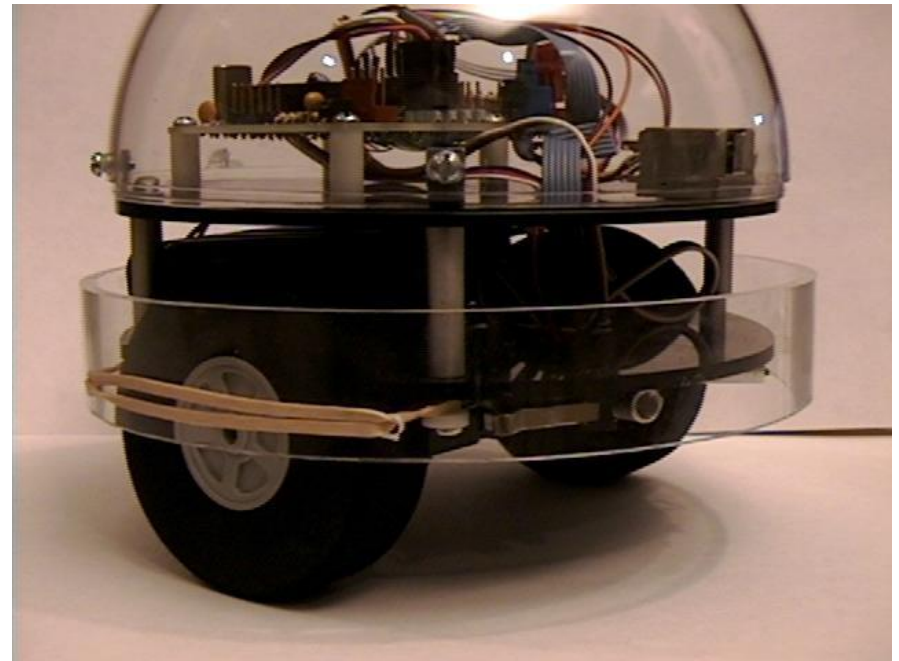
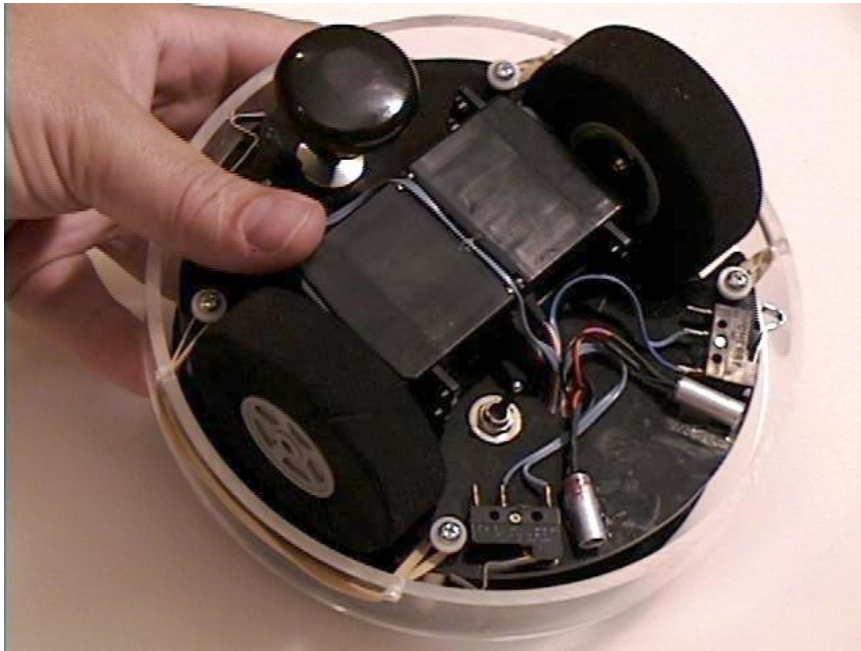
- Bumpers & Guards

- Impact/Collision sensor, senses pressure/contact
- Microswitches & wires or framework that moves
- Simple, cheap, binary output, easy to read

Feelers - Whiskers



Feelers - Bumpers & Guards



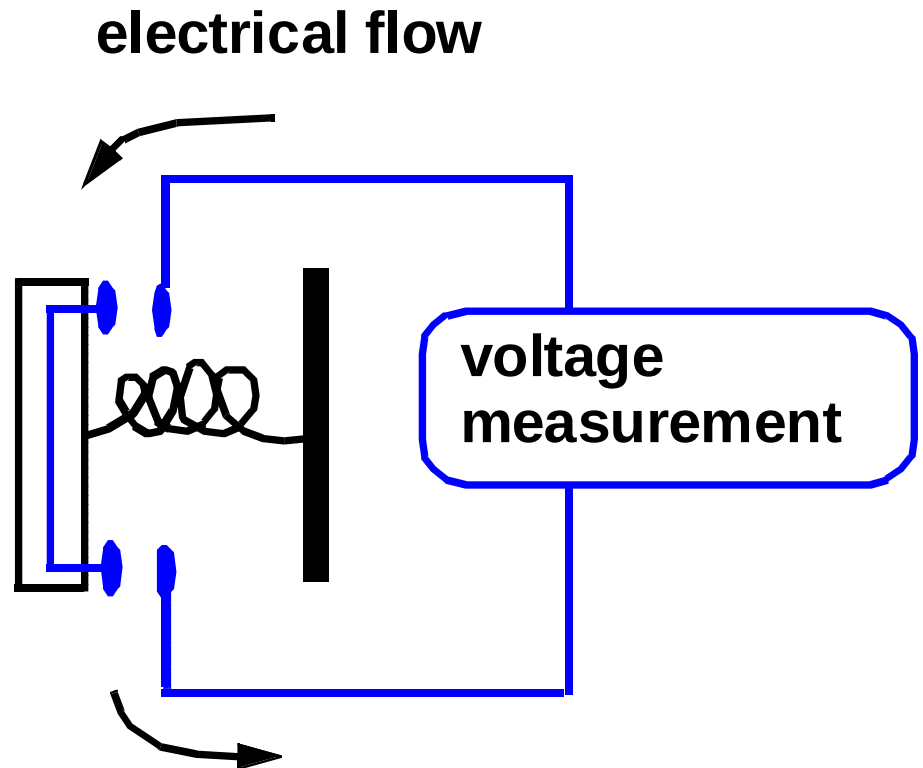
From Kevin Ross's "Getting Started Article (SRS Website)"

Touch sensors

**a simple
switch**



force
→



Sensors – Resistance

- Passive (sensor only)
 - Measures elec. resistance between objects
 - Measure sensor that varies resistance
 - Use absolute or differential readings
 - Other ideas?

Physics 101

Ohm's Law

$$V = I \times R$$

voltage

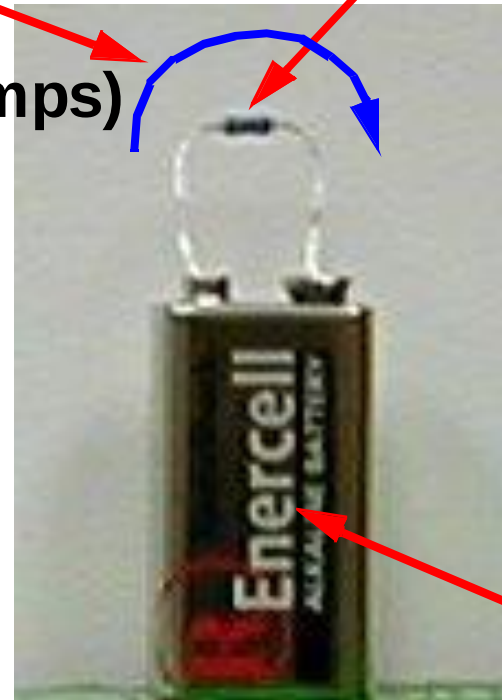
current

resistance

$$9 = 0.009 \times 1000$$

I
(0.009 Amps)

R (1000 Ohms)



V
(9 Volts)

Bend sensor

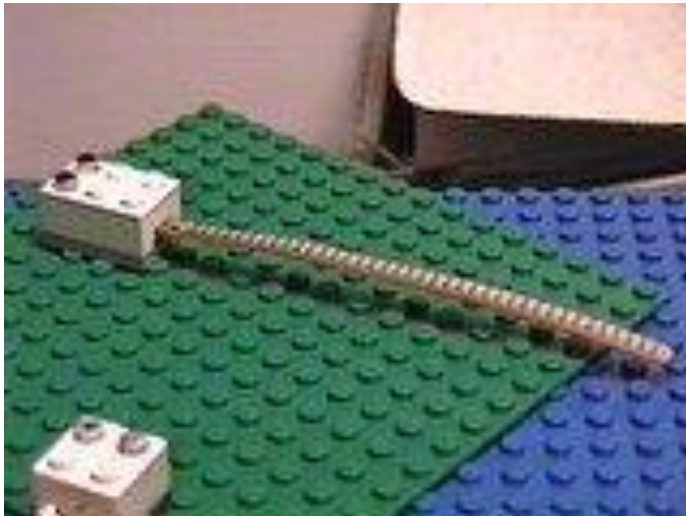
a variable resistor



resistance changes
as it bends

$$V = I \times R$$

assuming constant
current, the measured
voltage changes with
resistance



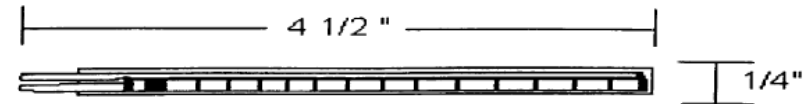
Flexion (bend) sensors

- resistance:
- 10 K Ω (0°);
- 30-40 K Ω (90°)



<http://www.flexpoint.com/>

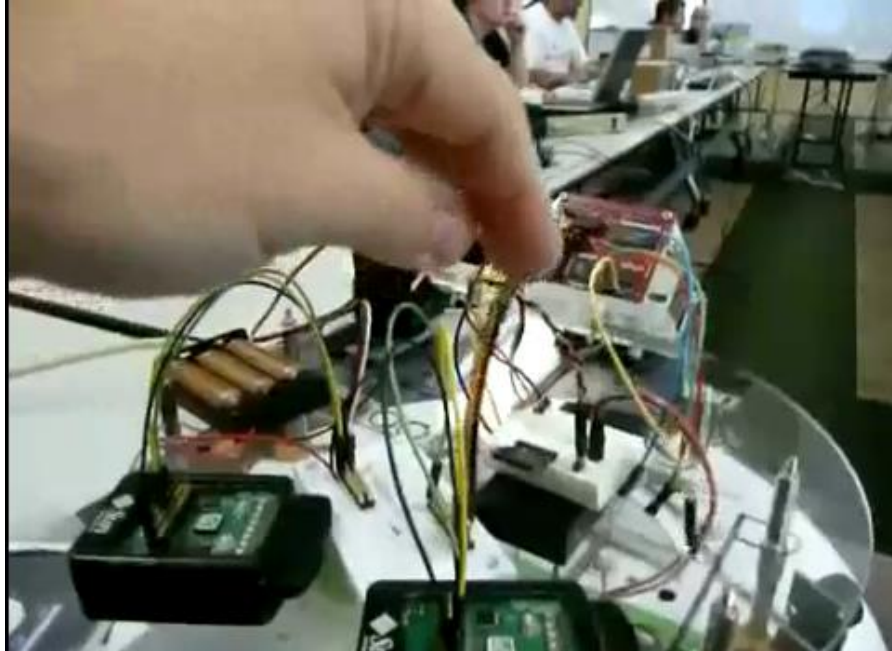
Flex Sensor



Nominal Resistance
Flex 0 Degrees: 10 K

Flex 90 Degrees
30-40 K

Felixon resistance Demo



<http://www.youtube.com/watch?v=m4E5SP7HCnk&feature=related>

Light sensor

photo-resistor

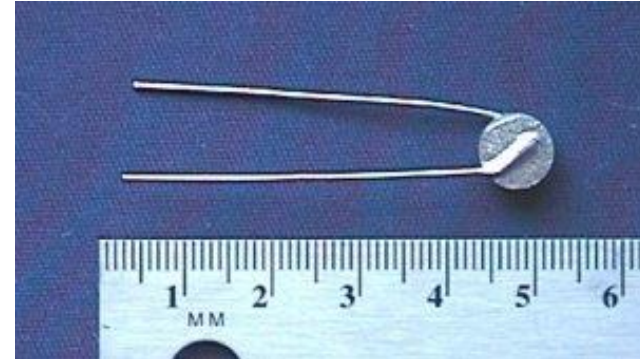


**resistance changes
with light intensity**



Temperature sensor

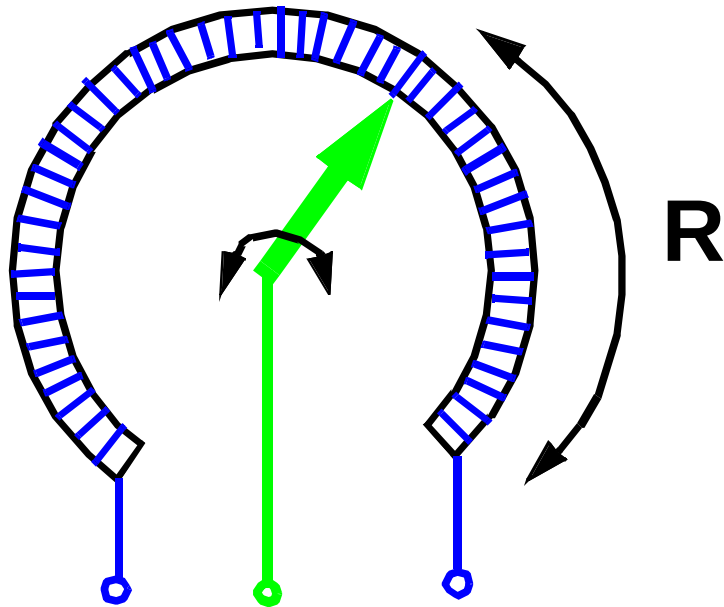
**thermal resistor
“thermistor”**



**resistance changes
with temperature**

Potentiometer

**another
rotational sensor**



**resistance changes
with position
of dial**

Sample problem



Bend sensor specs:

100 Ω when straight

1000 Ω when bent

$$V = I \times R$$

$$I = \frac{V}{R}$$

Given a 5 V source,
what is the min. and max.
current that is drawn?

$$\text{min} = \frac{5}{1000} = 5 \text{ mA}$$

$$\text{max} = \frac{5}{100} = 50 \text{ mA}$$

Sensors – Capacitive

- Passive
 - Really doesn't work (Needs excitation)
- Active (emitting)
 - Generate AC or DC voltage
 - Apply to external environment
 - Measure current to determine Resistance
 - Short range applications

Sensors - Capacitive

- Emit an electric field below the sensor.
- Nulled to a known “void” wall area.
- Detect capacitance difference due to underlying material density.



Sensors – Inductive

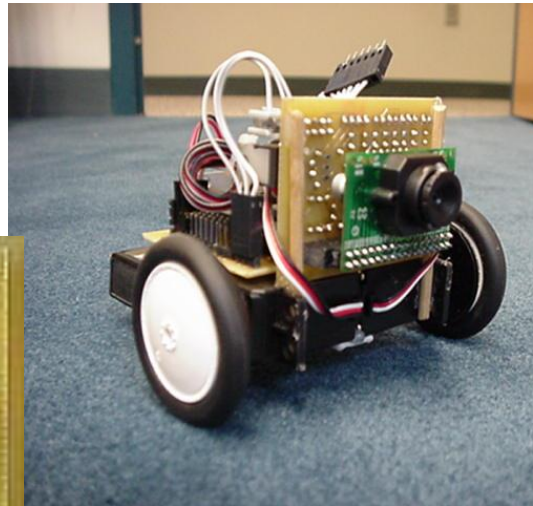
- Passive
 - Really doesn't work (Needs excitation)
- Active (emitting)
 - Current flows through inductor
 - Magnetic field mostly ignores non-metals
 - Inductance changes with metallic proximity
 - Short range applications

Sensors – Visual

- Active (emitting)
 - Camera with field of view illumination
 - Looks for particular reflections
 - Filter removes non-significant light sources
 - Linear array senses single axis of motion
- Passive (camera only)
 - Scans field of interest
 - Looks for objects, artifacts, features of interest
 - Processes digital data to simplified interpretation

Sensors – Visual

- CMUCam
- Linear Optical Array



Sensors – Color

- Active (emitting)
 - Selective field illumination (specific color(s))
 - Sensor filter removes extraneous light sources
 - Output can be analog (prop.) or digital (on/off)
- Passive (sensors only)
 - Different sensors for different colors
 - Color filter removes extraneous light sources
 - Output can be analog (prop.) or digital (on/off)

Sensors – Color

<http://robotroom.com/ColorSensor.html>

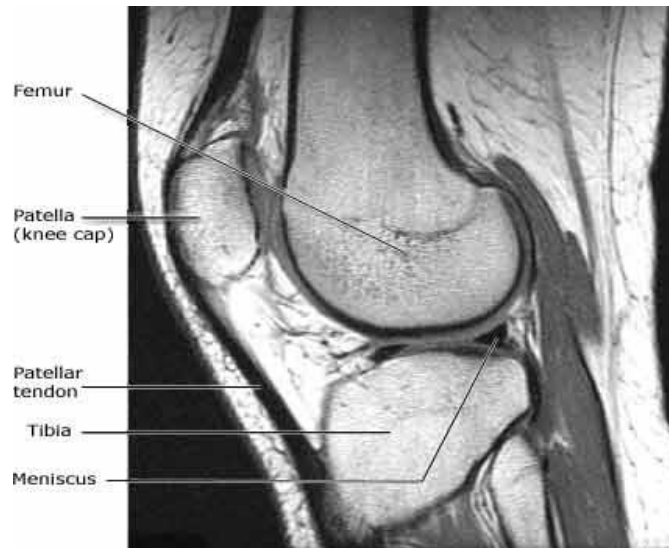


Sensors – Magnetic

- Active (emitting)
 - Metal detectors
 - Follows metallic strips on or under the floor
 - Magnetometer
 - Magnetic Resonance Imaging (MRI)
- Passive (sensors only)
 - Compass
 - Magnetic field sensor (→oscillating current)

Sensors – Magnetic

From HowStuffWorks.com & RadiologyInfo.org



Motion (Orientation/inclination sensors

- Acceleration
- Gyroscope
- Compass
- Tilt Sensor

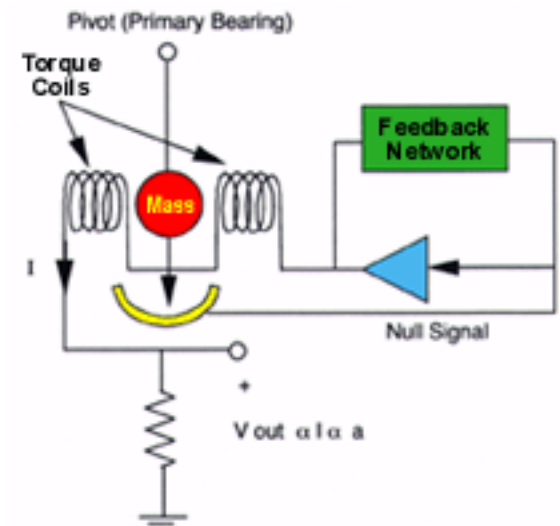
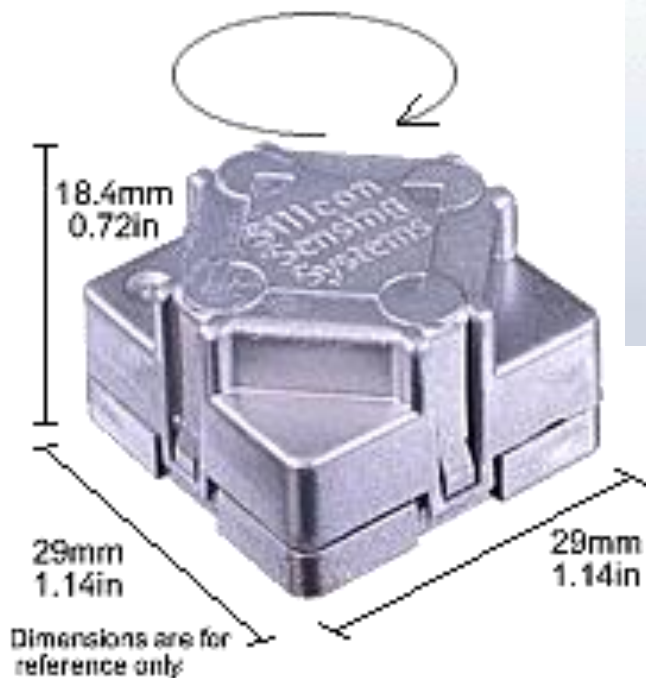
Sensors – Orientation

- Rate Gyros
 - Output proportional to angular rotation speed
 - Integrate to get position
 - Differentiate to get acceleration
- DC Accelerometer
 - Output proportional to sine of vertical angle

Sensors – Motion

Rate Gyro – Silicon Sensing Systems

Servo Accel – Sensorland.com



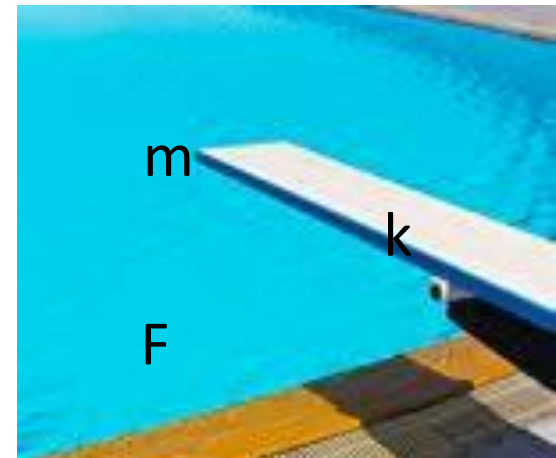
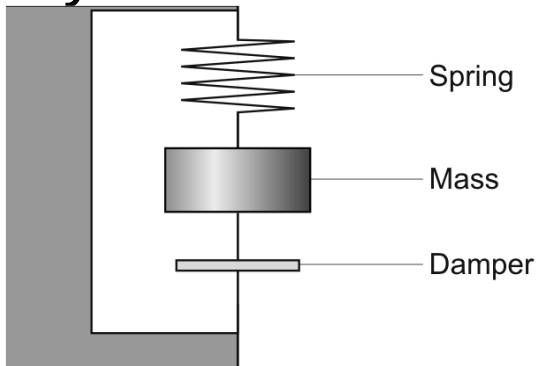
Accelerometer

<http://en.wikipedia.org/wiki/Accelerometer>

- Functions:
 - measure acceleration in one or more directions, position can be deduced by integration.
 - Orientation sensing : tilt sensor
 - Vibration sensing
- Methods:
 - Mass spring method ADXL78 (from Analog Device)
 - Air pocket method (MX2125)

Accelerometer

- $F = kx = ma$
- $a = \frac{kx}{m}$
- Accelerometers measure all external forces acting upon them, including gravity
- accelerometer acts like a spring–mass–damper system



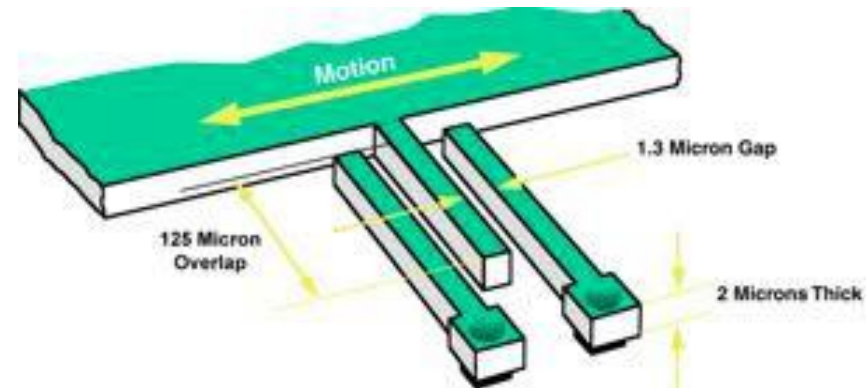
Accelerometers

Measure Linear Acceleration

- +Cheap
- Limited DoF
- Need to integrate for speed and distance
- Errors accumulate



- Modern accelerometers use Micro Electro-Mechanical Systems (MEMS) consisting of a spring-like structure with a proof mass. Damping results from the residual gas sealed in the device.
- Omnidirectional accelerometer by 3 accelerometers in 3 orthogonal directions



ADXL78 (from Analog Device

http://www.analog.com/UploadedFiles/Data_Sheets/ADXL78.pdf)

- Mass spring type (output acceleration in G)
- Measure the capacitance to create output

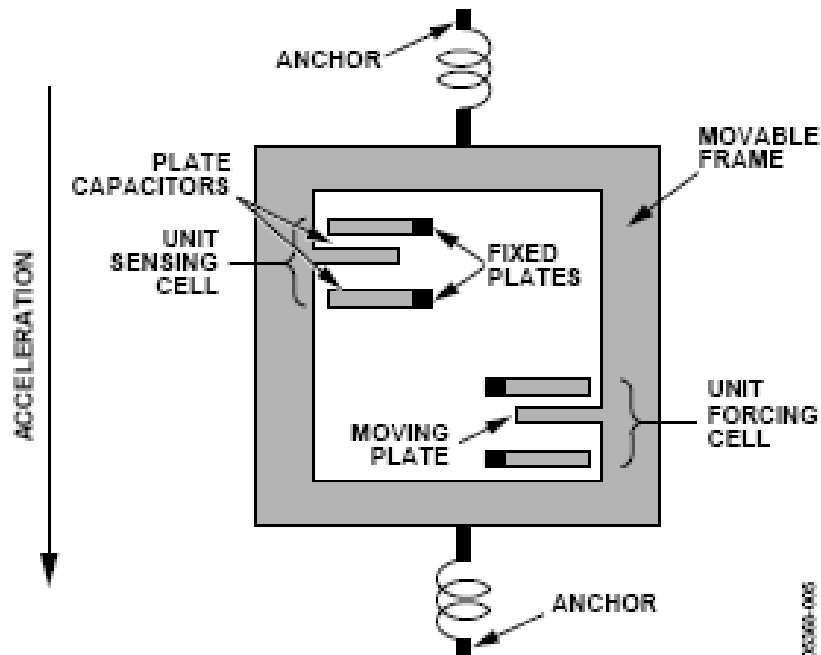


Figure 5. Simplified View of Sensor Under Acceleration

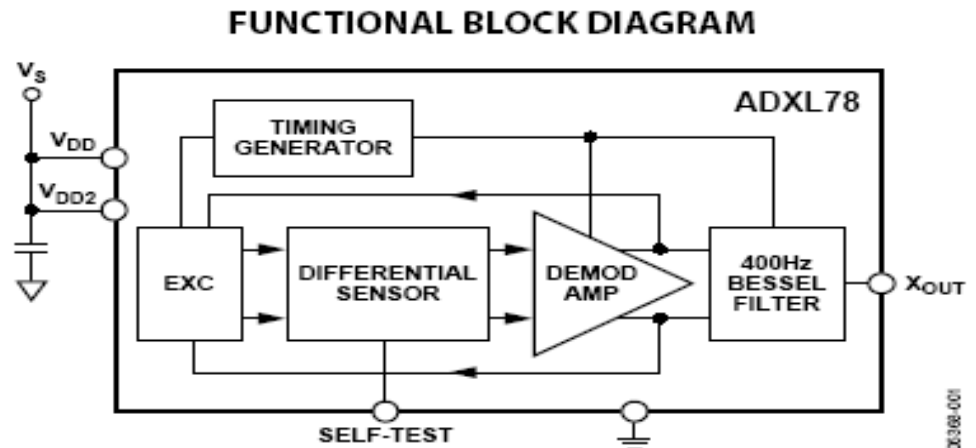


Figure 1.

ADXL330 accelerometer for measuring accelerations in three (X,Y,Z) directions

http://www.analog.com/UploadedFiles/Data_Sheets/ADXL330.pdf

- 3D accelerometer

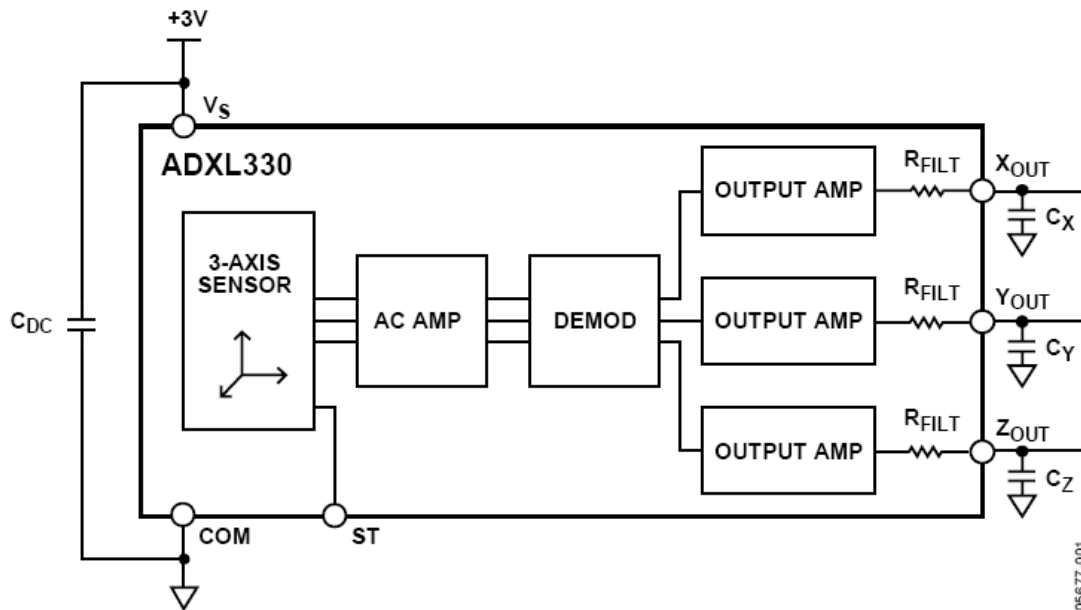


Figure 1.

$$\ddot{x} = \frac{dx^2}{dt^2} = \text{acceleration}$$

$$\dot{x} = \int \ddot{x} dt = \text{velocity}$$

$$x = \iint \ddot{x} dt = \text{distance}$$

Recover velocity and distance by integration
Problem: noise may be integrated

- Can also measure the direction of gravity

2D translational accelerometer MX2125 (from www.parallax.com)

- Gas pocket type
- When the sensor moves, the temperatures of the 4 sensors are used to evaluate the 2D accelerations

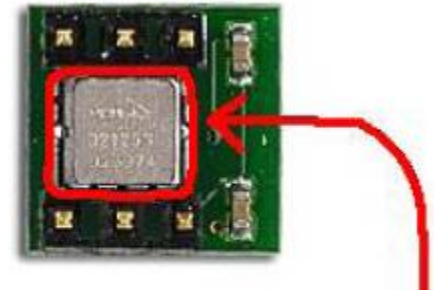
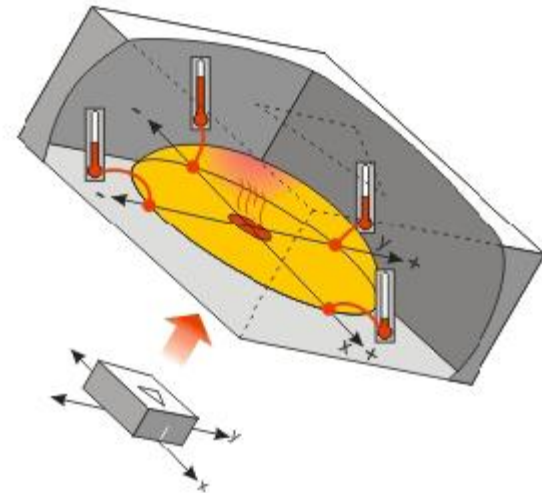
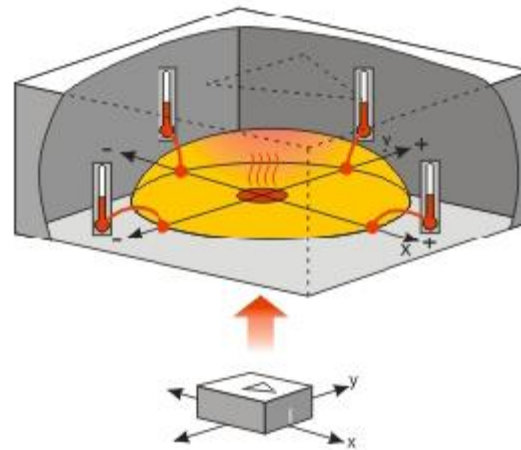
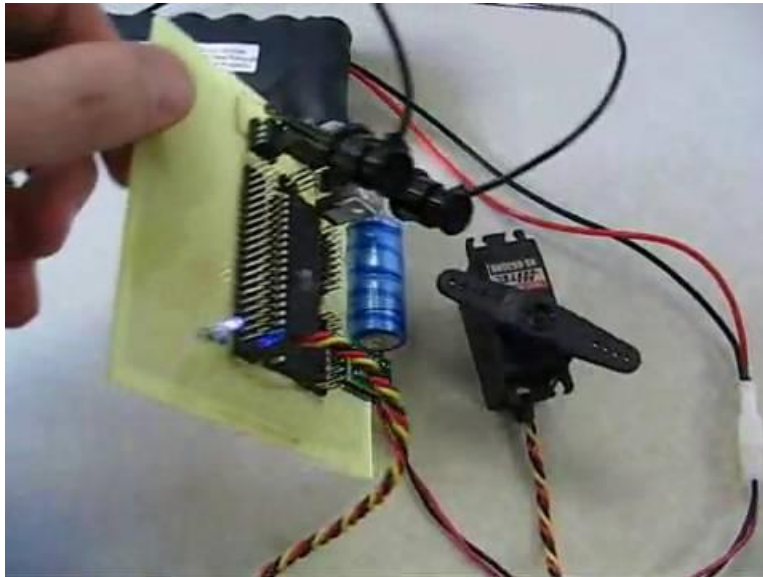


Figure 2 - Accelerometer Heated Gas Pocket



Accelerometer demo: orientation sensing

-



Sensor demo

<http://www.youtube.com/watch?v=9NEiBDBXFEQ>



Self-balance Robot

<https://www.youtube.com/watch?v=mwAK-fiPyk4>

Accelerometer demo :

Tilt sensing demo



Tilt sensing demo

<http://www.youtube.com/watch?v=C6uVrYz-j70>

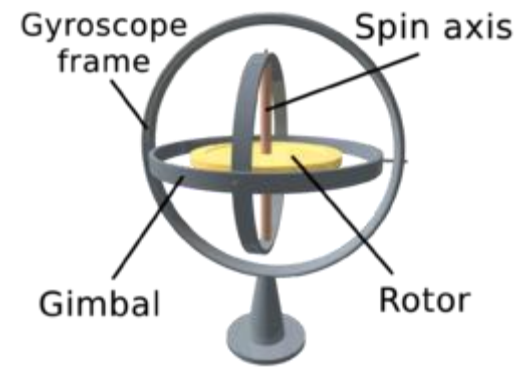


Application:
Xiaomi Mi band fitness

<https://www.youtube.com/watch?v=WvMrrQ0tMR4>

Gyroscopes

<http://en.wikipedia.org/wiki/Gyroscope>



Traditional
mechanical
Gyroscope

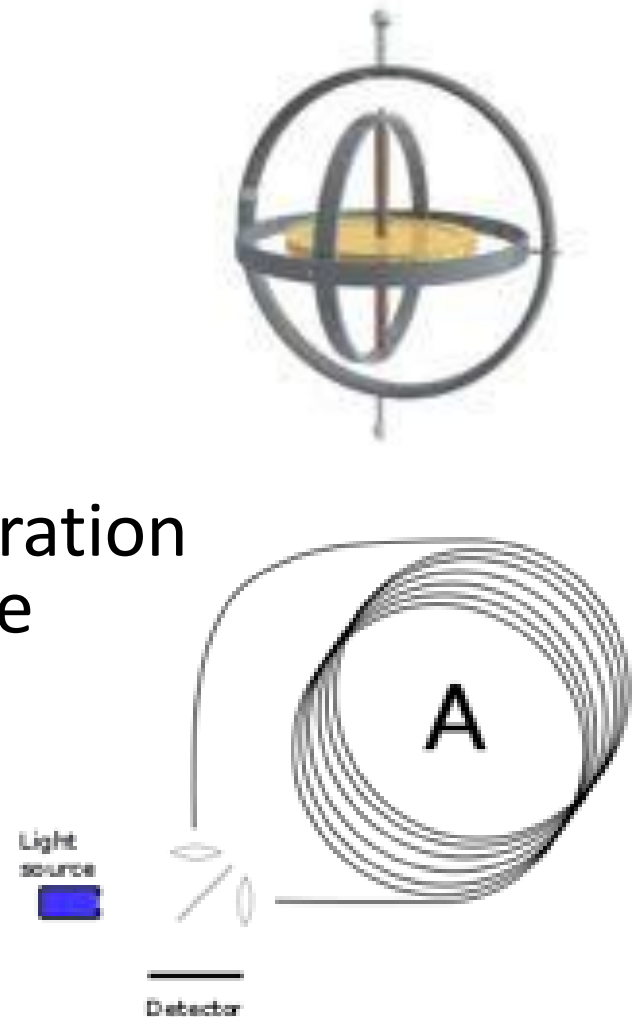
- Absolute Gyroscope measures: measures orientation
 - rotational angles θ_x , θ_y , θ_z
- Rate Gyroscope (or simply gyroscope)
 - measures the rate of rotation along 3-axes of X (pitch), Y (roll), and Z (yaw).
 - Meaning Measure $d\theta_x/dt$, $d\theta_y/dt$, $d\theta_z/dt$
 - Also need integration to recover rotational angles
 - Modern implementations are using Microelectromechanical systems (MEMS) technologies.

Gyroscope

- Heading sensors that preserve their orientation in relation to a fixed reference frame
 - absolute measure for the heading of a mobile system.
- Two categories, the mechanical and the optical gyroscopes
 - Mechanical Gyroscopes
 - Standard gyro (angle)
 - Rate gyro (speed)
 - Optical Gyroscopes
 - Rate gyro (speed)

Gyroscopes

- Measures orientation
- Very expensive, infeasible to miniaturize
- *Rate* gyroscopes measure rotational speed (variation)
- Implemented using MEMS vibration devices, measure Coriolis force
- Applications
 - Correct heading



Gyroscope example

ADXRS401

http://www.analog.com/UploadedFiles/Data_Sheets/ADXRS401.pdf

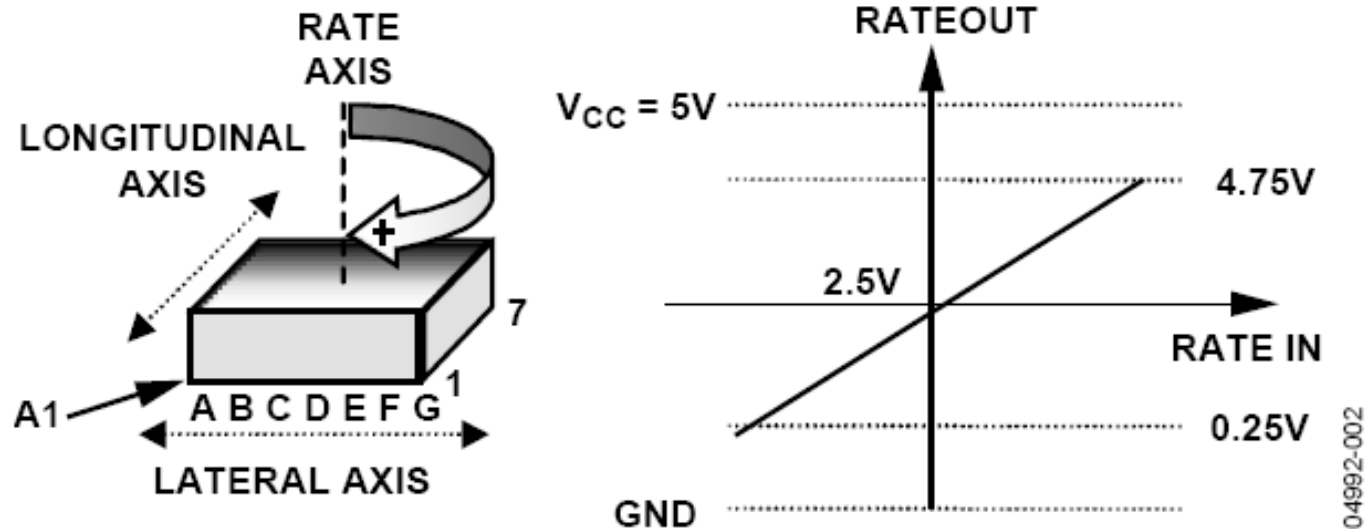


Figure 2. RATEOUT Signal Increases with Clockwise Rotation

- **FEATURE**
 - Complete rate gyroscope on a single chip Microelectromechanical systems (MEMS)
 - Z-axis (yaw-rate) response
- **APPLICATIONS**
 - GPS navigation systems
 - Image stabilization
 - Inertial measurement units
 - Platform stabilization

●



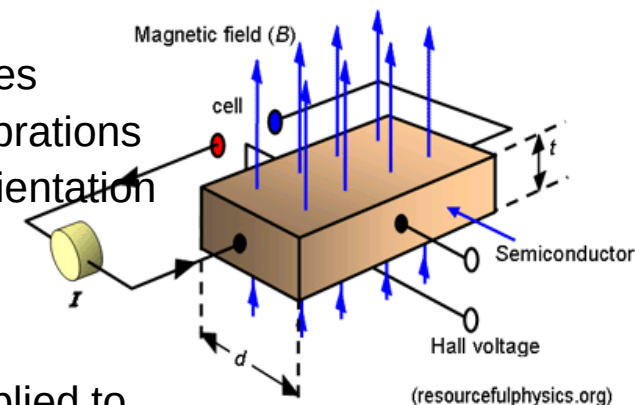
Sensors (v.5e)

Inertial measurement unit (IMU)

- A device that uses gyroscopes and accelerometers to estimate the relative positions, and six degrees of velocities (translation and rotation)

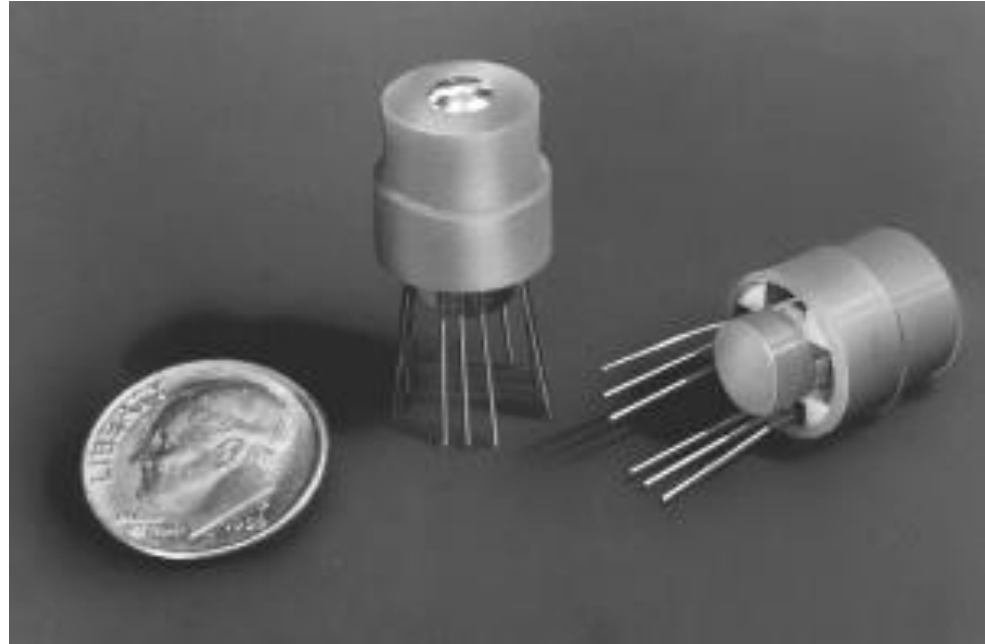
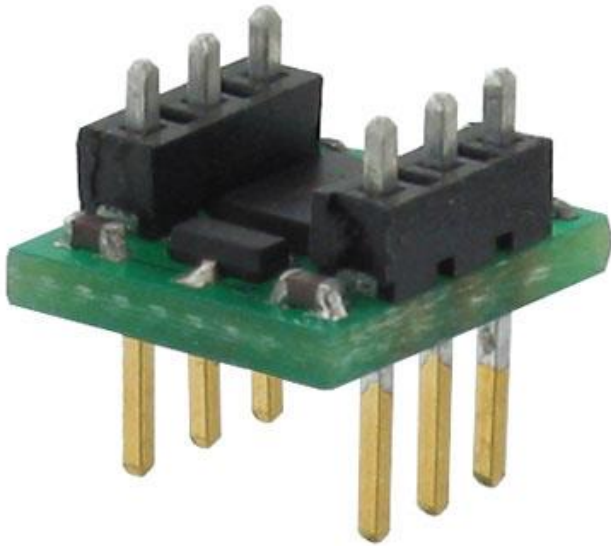
Compass

- Since over 2000 B.C.
 - when Chinese suspended a piece of naturally magnetite from a silk thread and used it to guide a chariot over land.
- Magnetic field on earth
 - absolute measure for orientation (even birds use it for migrations (2001 discovery))
- Large variety of solutions to measure the earth magnetic field
 - mechanical magnetic compass
 - direct measure of the magnetic field (Hall-effect, magneto-resistive sensors)
- Major drawback
 - weakness of the earth field ($30 \mu\text{Tesla}$)
 - easily disturbed by magnetic objects or other sources
 - bandwidth limitations (0.5 Hz) and susceptible to vibrations
 - not feasible for indoor environments for absolute orientation
 - useful indoor (only locally)



Hall effect: semiconductor in magnetic field. When current is applied to semiconductor across its length, there will be a voltage difference in the perpendicular direction, depending on the orientation of the magnetic field.

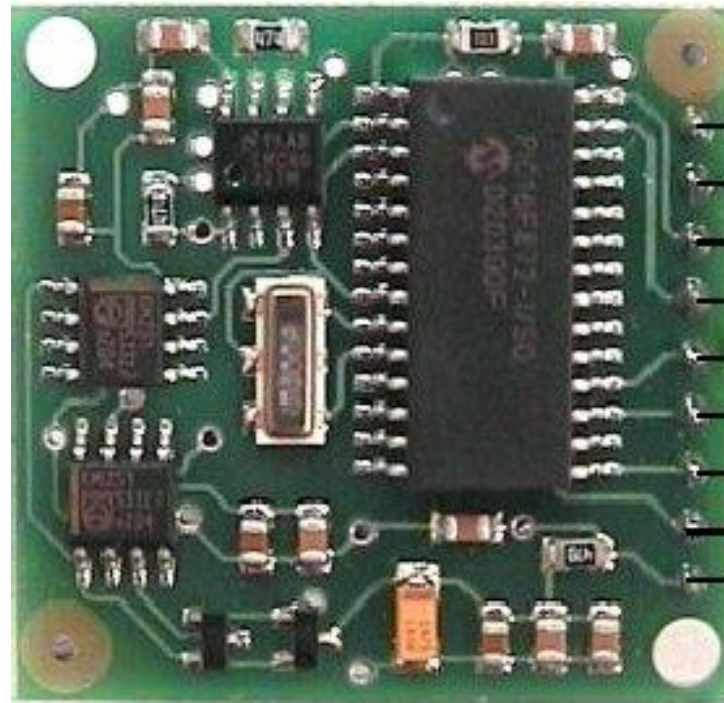
Sensors – Compass (Orientation)



- Track bearing & distance to determine position
- L: Parallax.com, \$30
- R: Dinsmoresensors.com, \$13-\$37

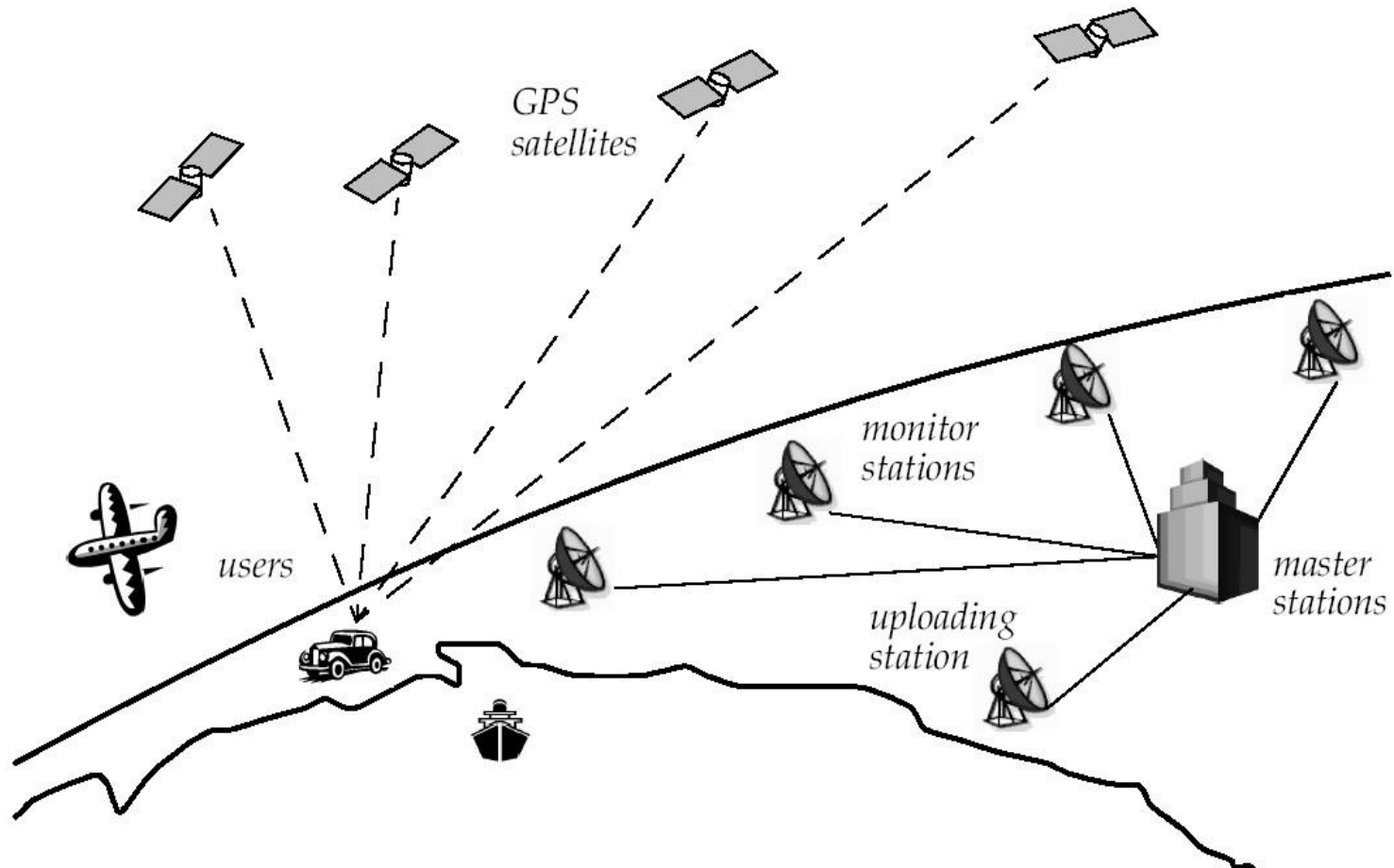
Compass-- the Philips KMZ51 magnetic field sensor

- 50/60Hz (high) operation, a jitter (noise) of around 1.5°



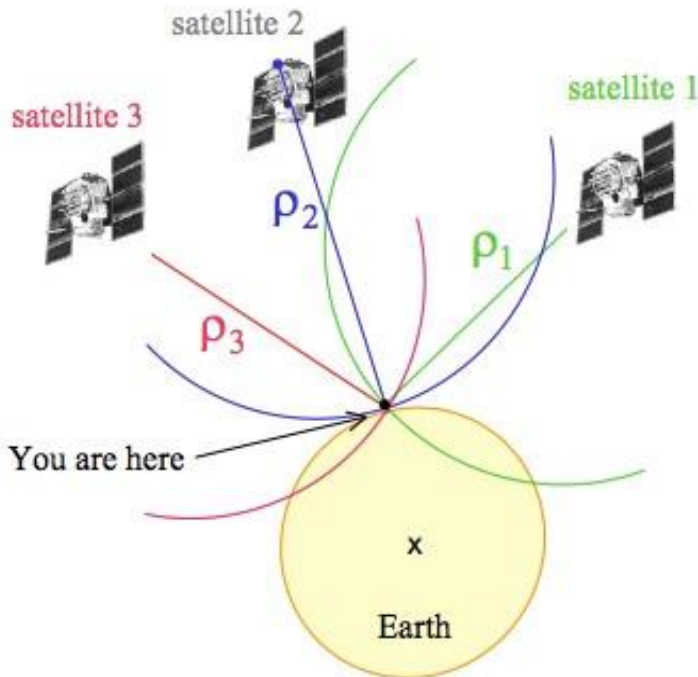
- Pin 9 - 0v Ground
- Pin 8 - No Connect
- Pin 7 - 50/60Hz
- Pin 6 - Calibrate
- Pin 5 - No Connect
- Pin 4 - PWM
- Pin 3 - SDA
- Pin 2 - SCL
- Pin 1 - +5v

Global Positioning System (GPS) (2)



GPS positioning

- Simple positioning principle
- Satellites send signals, receivers received them with delay



$$\rho = (t_r - t_e) \times \text{speed of light}$$

$$\rho = \sqrt{(X_s - X_r)^2 + (Y_s - Y_r)^2 + (Z_s - Z_r)^2}$$

If we know at least three distance measurements, we can solve for position on earth.

In practice four are used since the time difference between the GPS receiver's clock and the synchronized clocks of the satellites is unknown.

General principle: beacon based sensors :

- indoor-GPS solutions, either active or passive beacons that are mounted in the environment at known locations.
- Passive beacons: infrared reflecting stickers arranged in a certain pattern ; 2D barcodes detected using cameras.
- Active beacons usually emit radio, ultrasound or a combination thereof, to estimate the robot's range to these beacons.

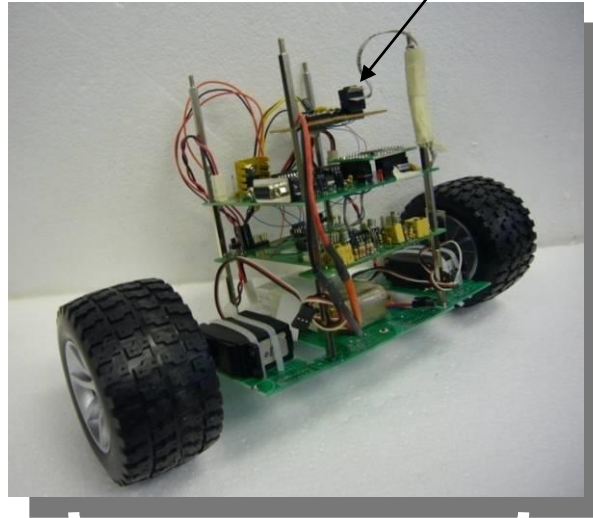
LATER ON...

Application of motion sensors

Self balancing robot

- by Kelvin Ko, KH wong
- <https://www.youtube.com/watch?v=dQWATsLa30g>

Motion sensors:
gyroscope and
accelerometer



20cm

35cm

Complementary filter

- Since

Gyroscope :
Accurate in **High**
frequency
measurement

Accelerometer:
Accurate in **low**
frequency
measurement

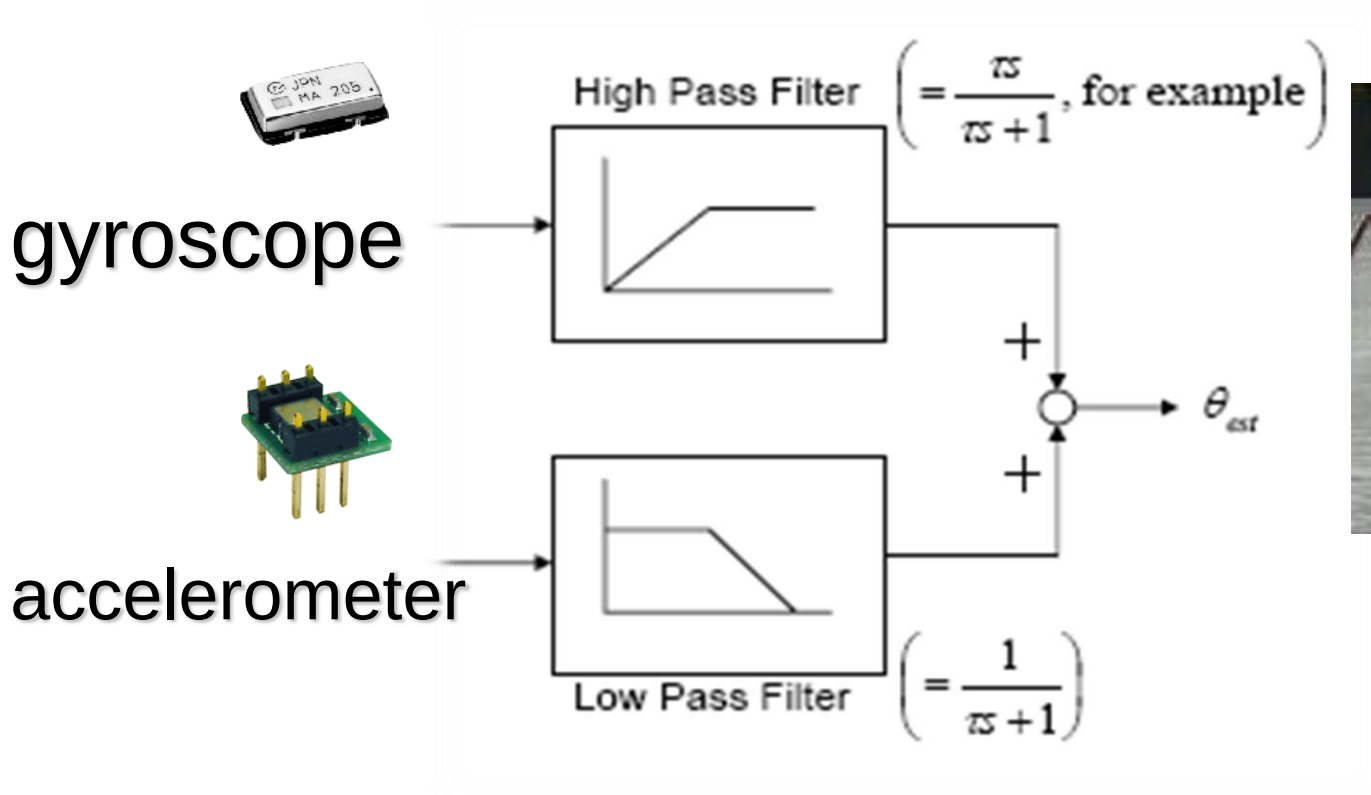
- Combine two sensors to find output at all frequencies

Complementary filter

θ =rotation angle, τ =filter time constant, s =laplace operator

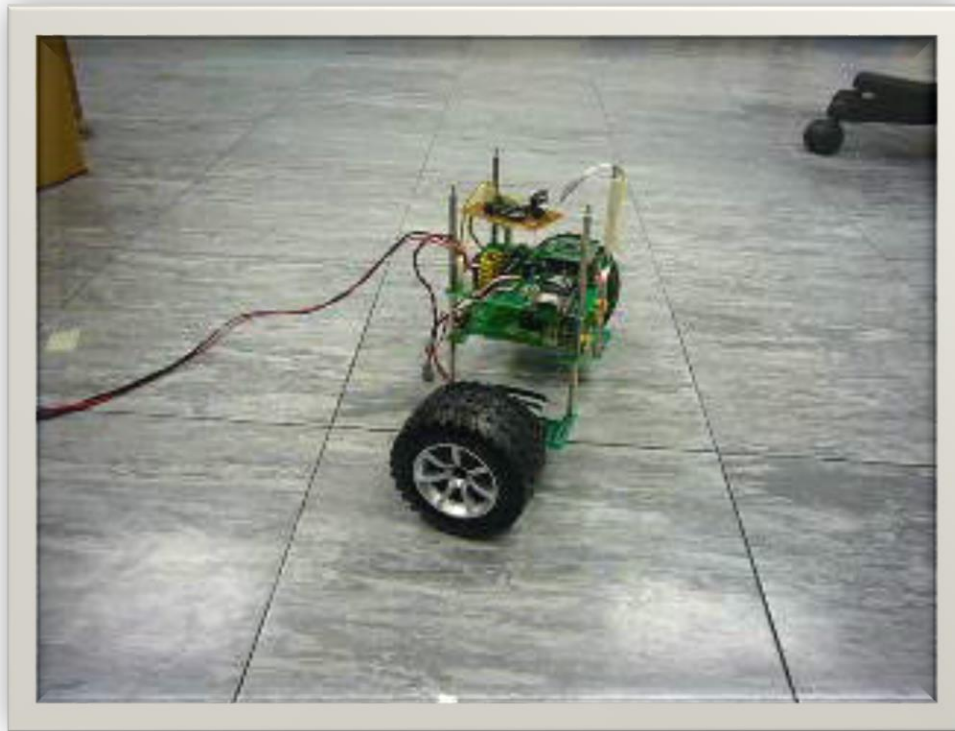
http://en.wikipedia.org/wiki/Low-pass_filter

http://en.wikipedia.org/wiki/High-pass_filter



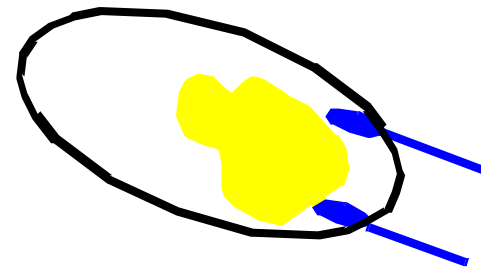
Self Balanced robot using complementary filter

<https://www.youtube.com/watch?v=dQWATsLa30g>



Tilt sensors

**another simple
switch**

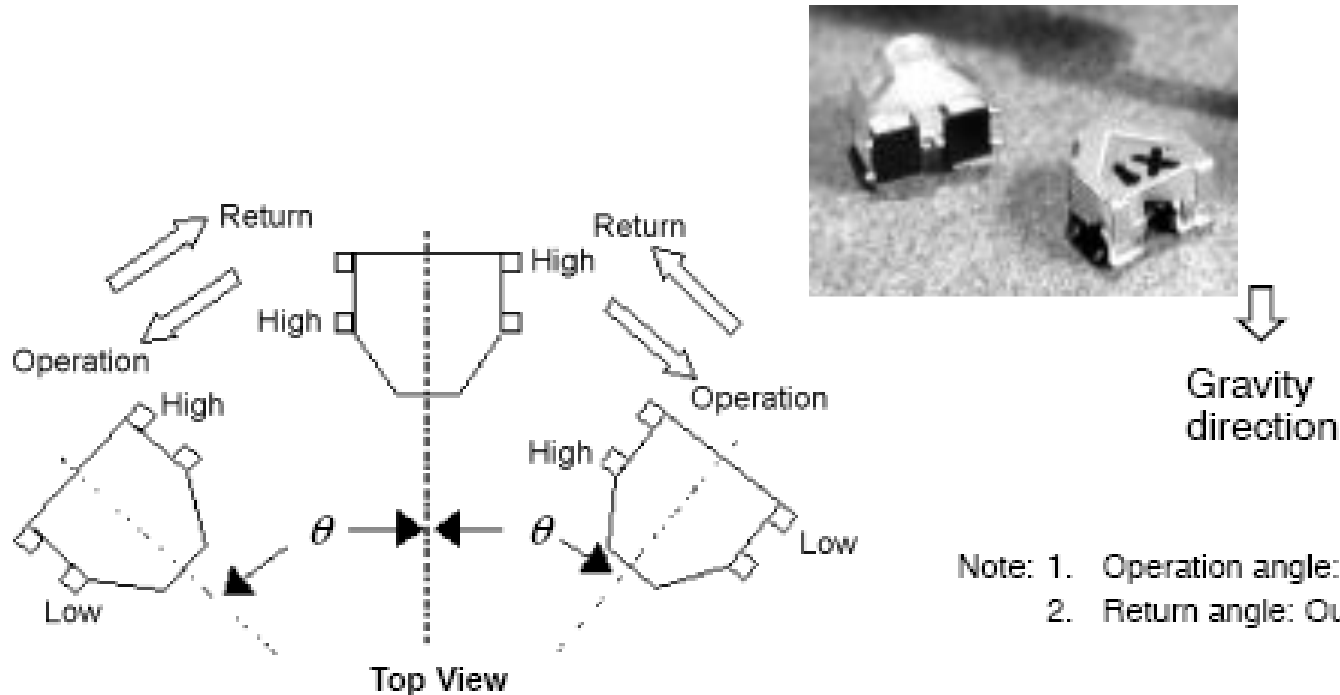


gravity

Tilt Sensor by OMRON

<http://rocky.digikey.com/WebLib/Omron%20Web%20Data/D6B.pdf>

- Detect tilting 35 ~ 65 degrees in right-and-left inclination



Force/pressure/strain

- Force-sensitive resistor (FSR)
- Strain gauge
- Flexion
- Air pressure

Force Sensing Resistors

<http://www.interlinkelectronics.com/library/media/papers/pdf/fsrguide.pdf>

f

- FSR402

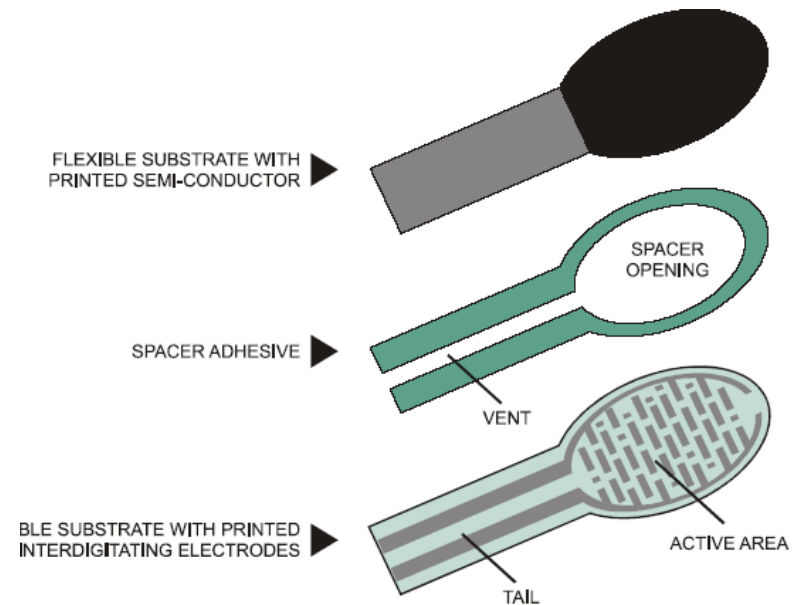
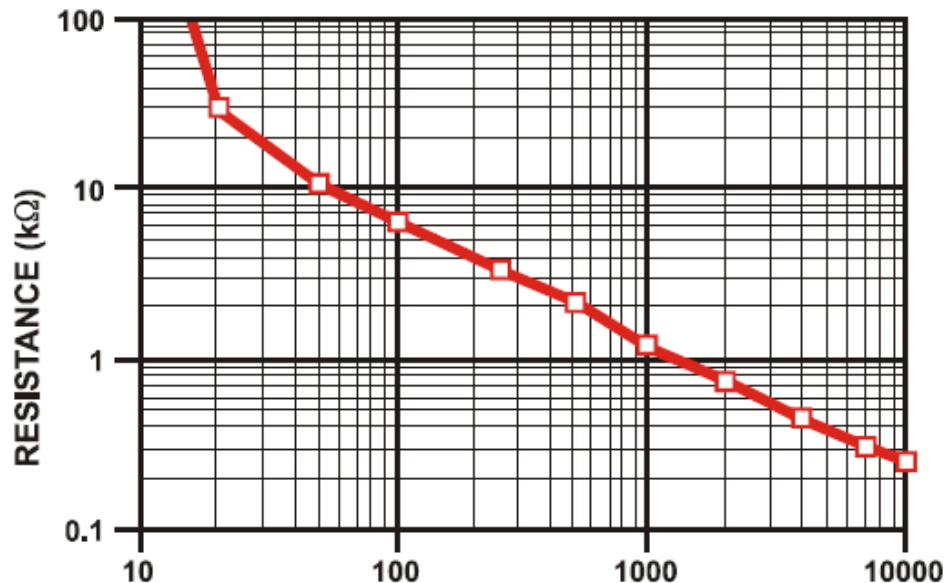
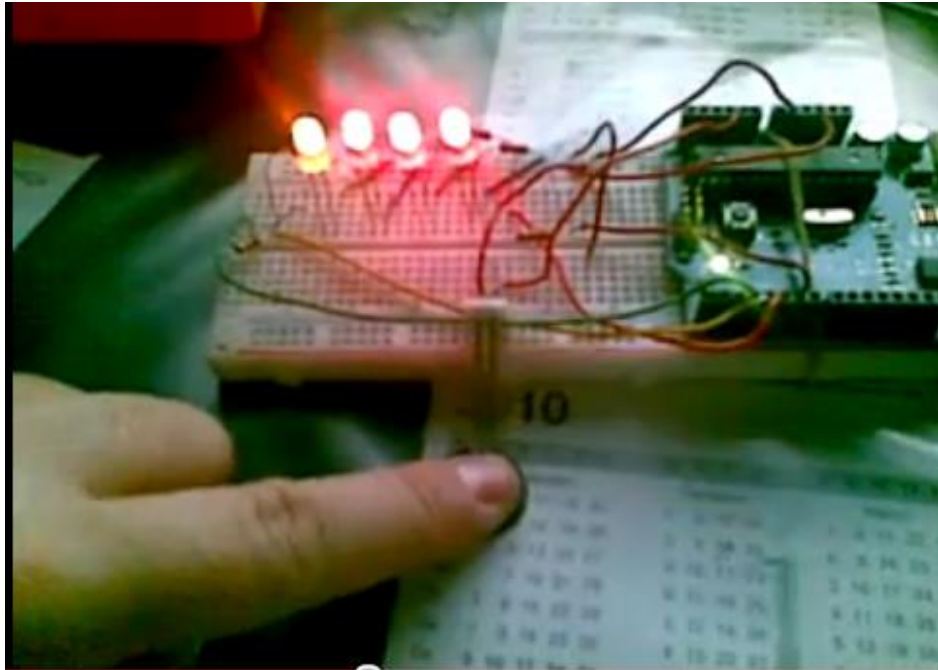


Figure 1: FSR Construction

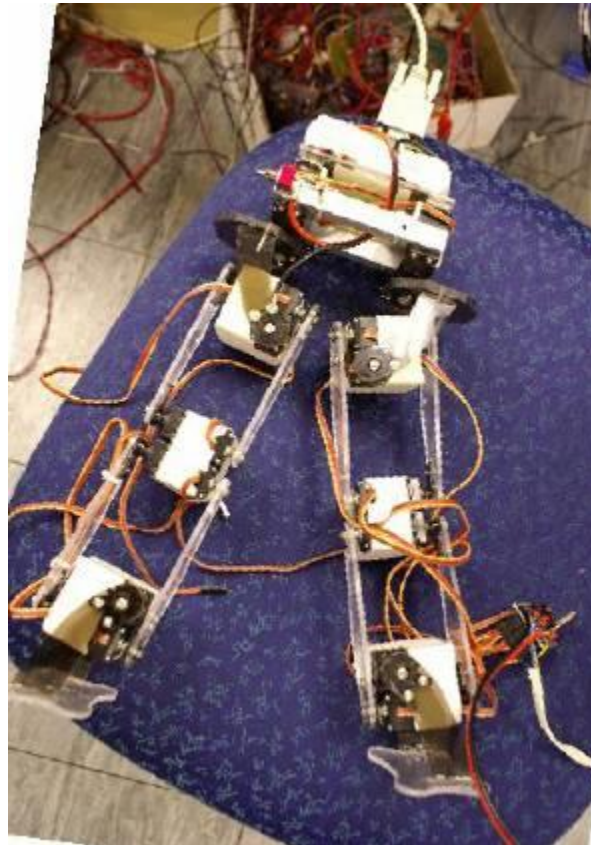
Force Sensing Resistor Demo



<http://www.youtube.com/watch?v=LQ21lXr6egs>

Application for a walking robot

- Walking robot



Application of force sensing resistance sensors to balance a walking robot

- Balancing

Neutral position



Floor tilted left
upper leg bend right

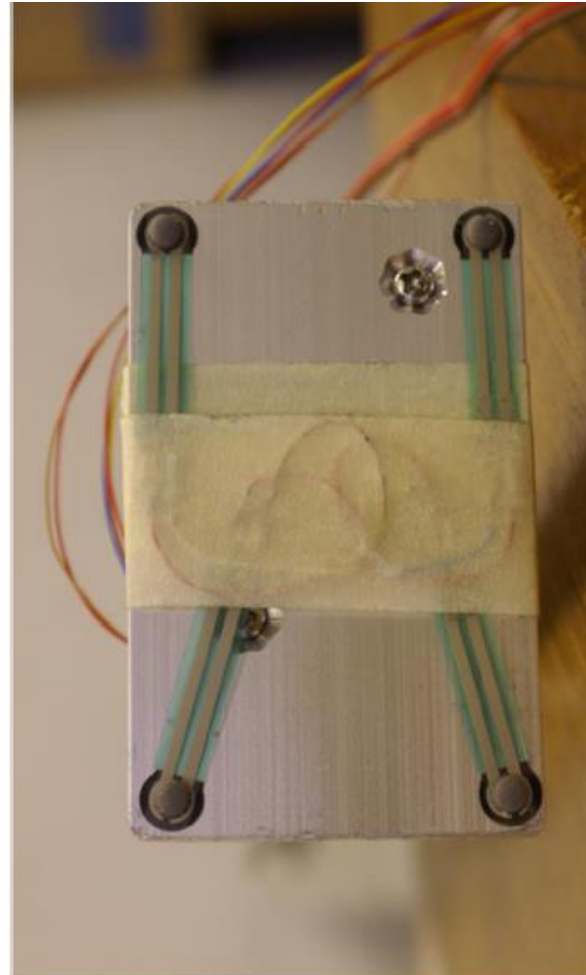
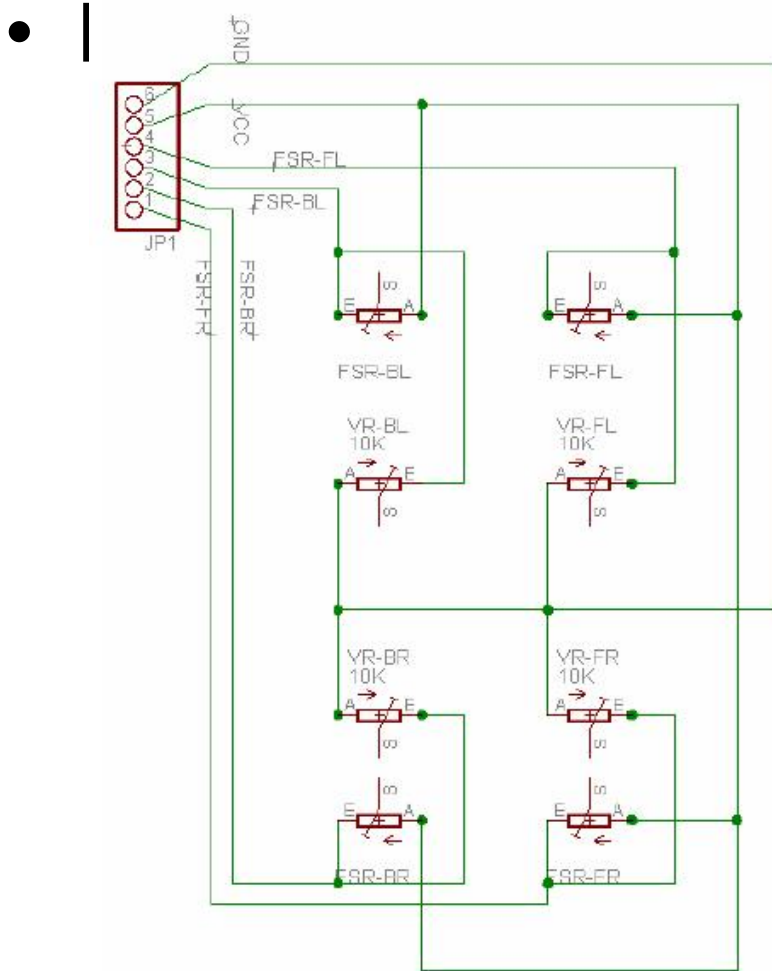


Floor tilted right
upper leg bend left



Four sensors under the foot

Four Force sensors under the foot



The Nao robot uses force feedback at its feet [http://en.wikipedia.org/wiki/Nao_\(robot\)](http://en.wikipedia.org/wiki/Nao_(robot))

•



<http://www.youtube.com/watch?v=2STTNYNF4lk>

<http://www.robotshop.com/Images/xbig/fr/robot-humanoide-nao-edition-academique-v3plus-aldebaran.jpg>

Strain Gauge : Force sensors

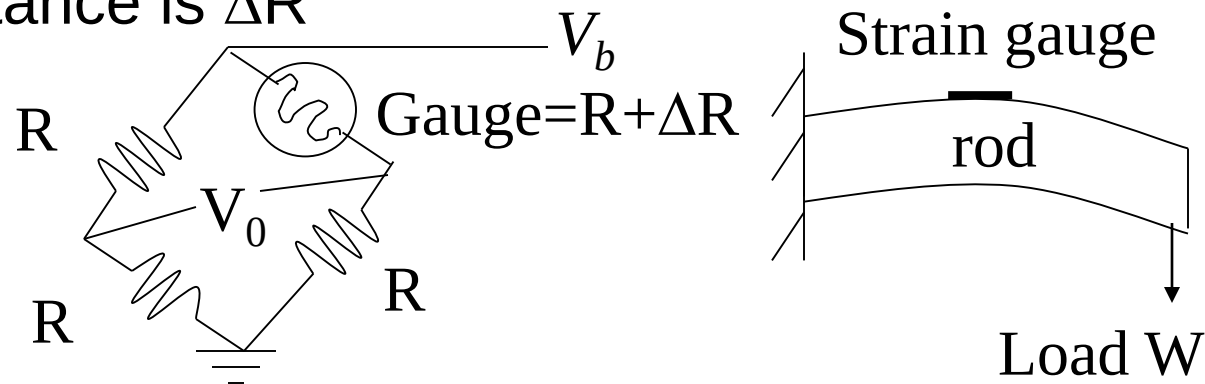
<http://www.meas-spec.com/myMeas/default/index.asp>

- Piezoelectric crystal: produces a voltage that is proportional to the force applied
- Strain gauge: cemented on a rod. One end of the rod is fixed, force is applied to the other end. The resistance of the gauge will change with the force.

https://en.wikipedia.org/wiki/Strain_gauge

Single element strain gauge (sensitive to temperature change.)

- Originally the length of the strain gauge is L , and the resistance is R when there is no load. When ΔW is applied to the rod, the change of the length of the strain gauge is ΔL , and the corresponding change of the resistance is ΔR



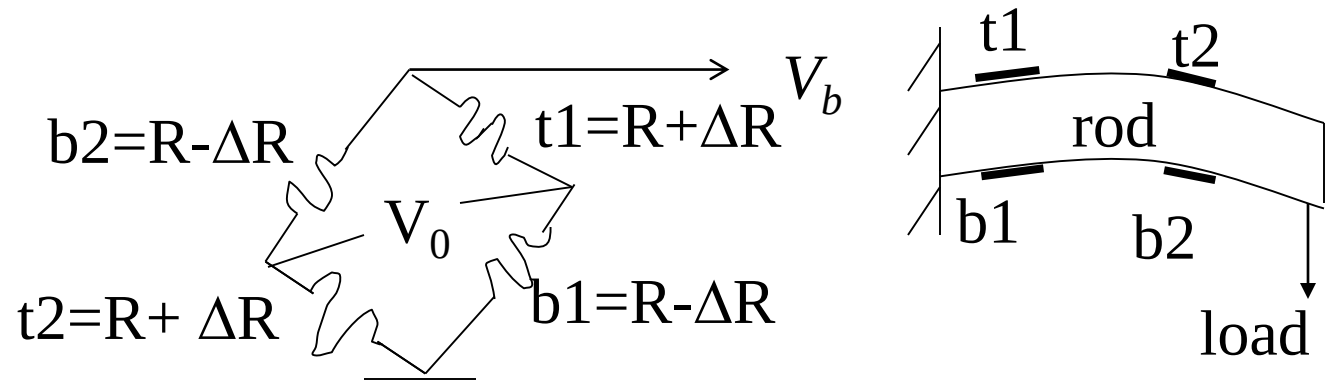
$$V_0 = V_b \left(\frac{R}{2R} - \frac{R}{2R + \Delta R} \right) = V_b \left(\frac{\Delta R}{4R + 2\Delta R} \right) \approx V_b \left(\frac{\Delta R}{4R} \right) = V_b \left(\frac{G}{4} \frac{\Delta L}{L} \right)$$

for $\frac{\Delta R}{R} = G \frac{\Delta L}{L}$ and G = strain gauge factor, L = length of the gauge

R = unstrained gauge resistance

Four-element (Wheatstone bridge) strain gauge sensor,

- Advantages : Four times more sensitive than single gauge system; not sensitive to temperature change.
- All gauges have unstrained resistance R .



$$V_0 = V_b \left(\frac{R + \Delta R}{R + \Delta R + R - \Delta R} - \frac{R - \Delta R}{R + \Delta R + R - \Delta R} \right) = V_b \left(\frac{2\Delta R}{2R} \right) = V_b \left(G \frac{\Delta L}{L} \right)$$

Sensors – Position/Location

- Wheel Encoders
 - Relative position & motion
 - Integrate/Differentiate for other parameters
- Global Positioning System
 - Absolute position/location on earth
 - Local differential error correction
 - Integrate/Differentiate for other parameters

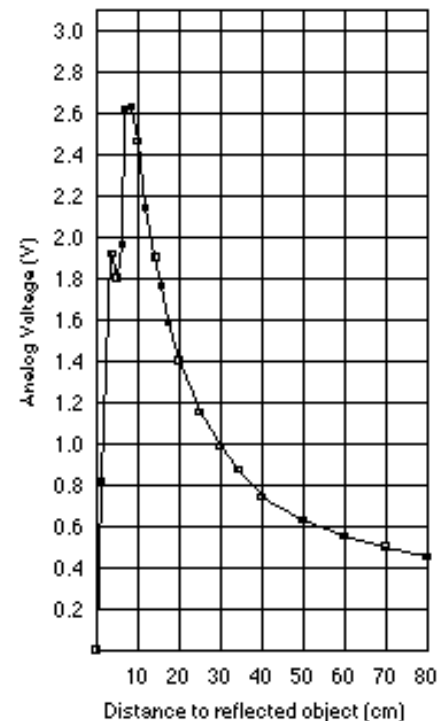
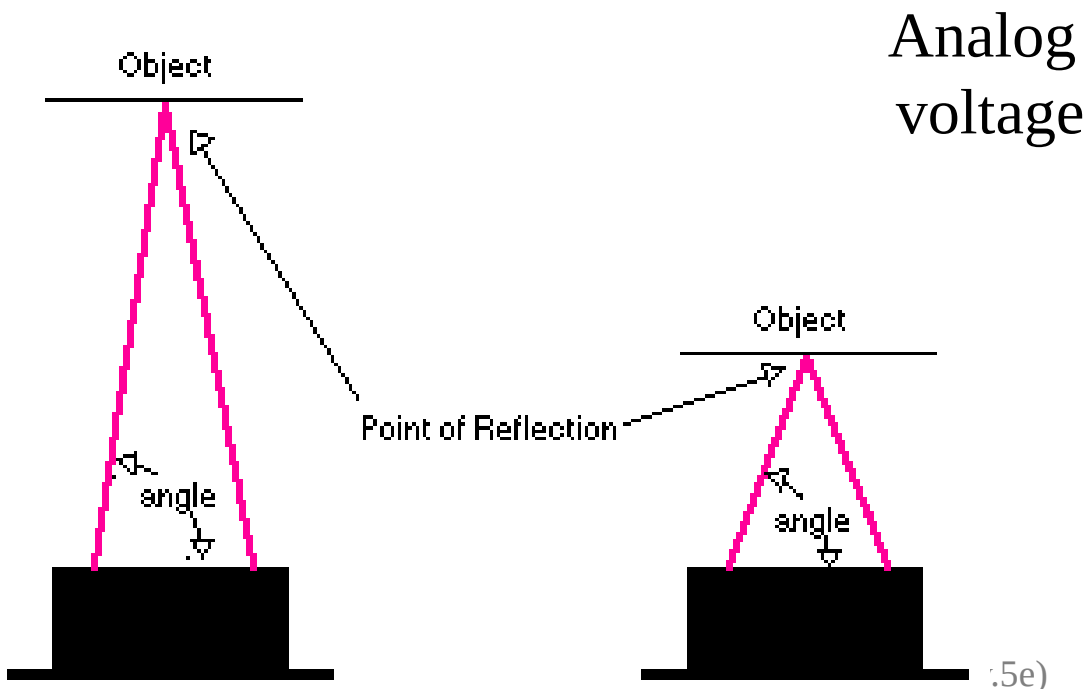
Position sensors

- Infra-red range sensor
- Linear and Rotary position sensors

Infra-red Range detectors by SHARP (4 to 30cm)

<http://www.acroname.com/robotics/info/articles/sharp/sharp.html>

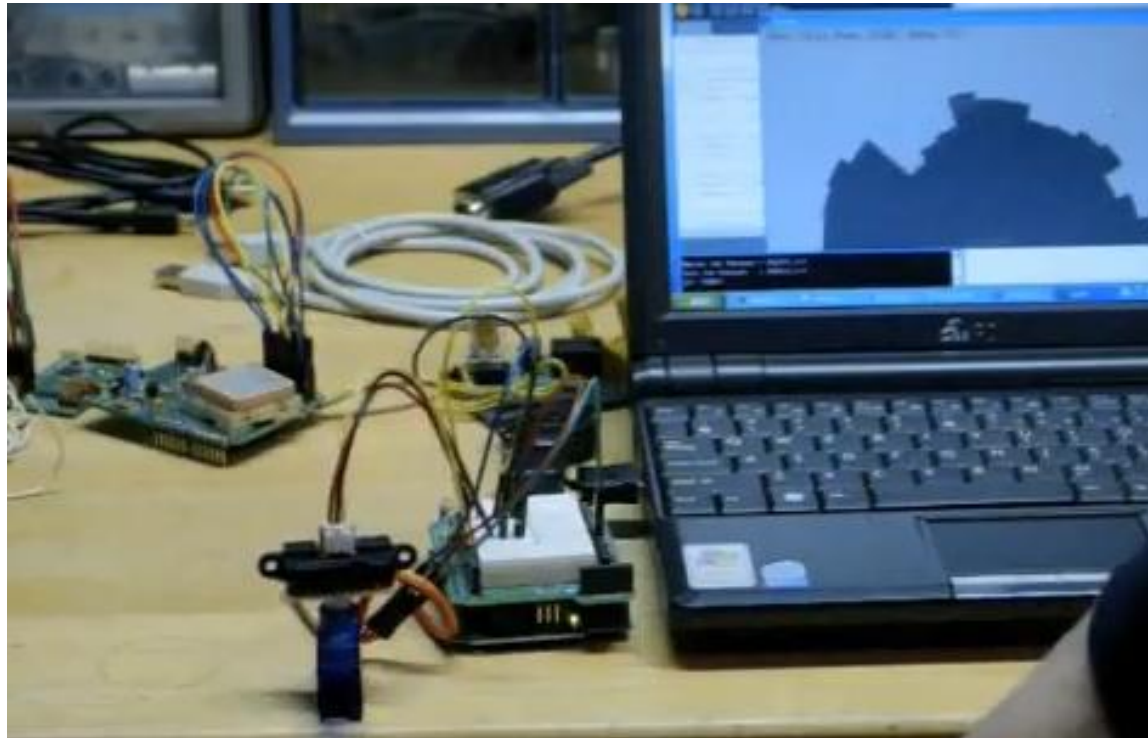
- An emitter sends out light pulses. A small linear CCD array receives reflected light.
- The distance corresponds to the triangle formed.



Distance to reflected object (cm) 92

IR radar using the Sharp range detector

-



<http://www.youtube.com/watch?v=tStBLAiQaC8&feature=related>

Encoders

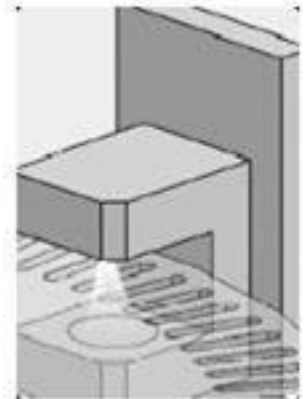
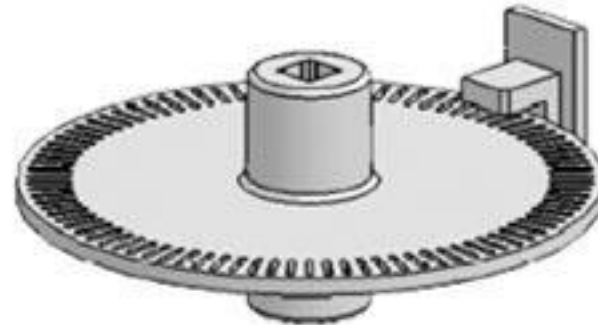
- ❑ Encoders measure rotational motion.
- ❑ They can be used to measure the rotation of a wheel.
- ❑ **Servo motors:** Used in conjunction with an electric motor to measure the motor's position and, in turn, control its position.

Robot Proprioception

Odometry/shaft encoders

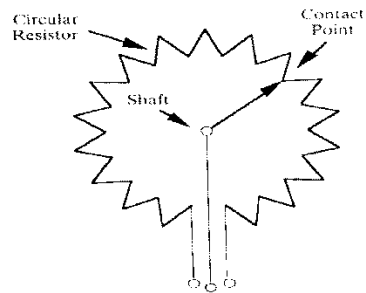
Light/Magnetic/Radar

- +Simple
- +Widely Available
- Limited Speed
- Slippage

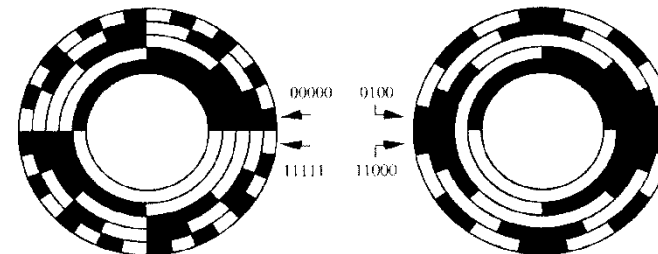


Position sensors

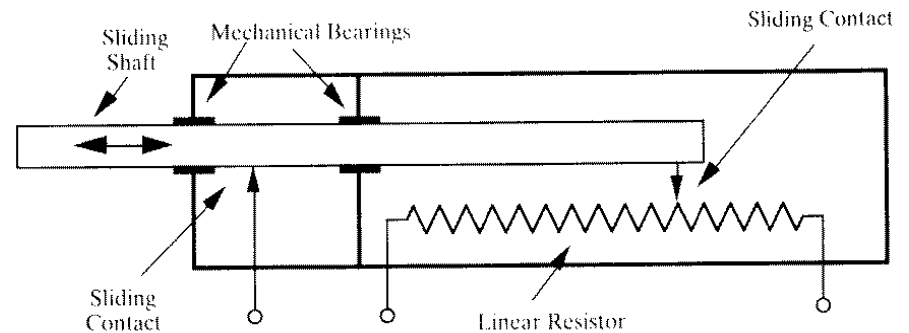
- Rotary



- Optical shaft encoder



Linear



Magnetic rotary encoder

(<http://www.renishaw.com/UserFiles/acrobat/UKEnglish/L-9517-9147.pdf>)

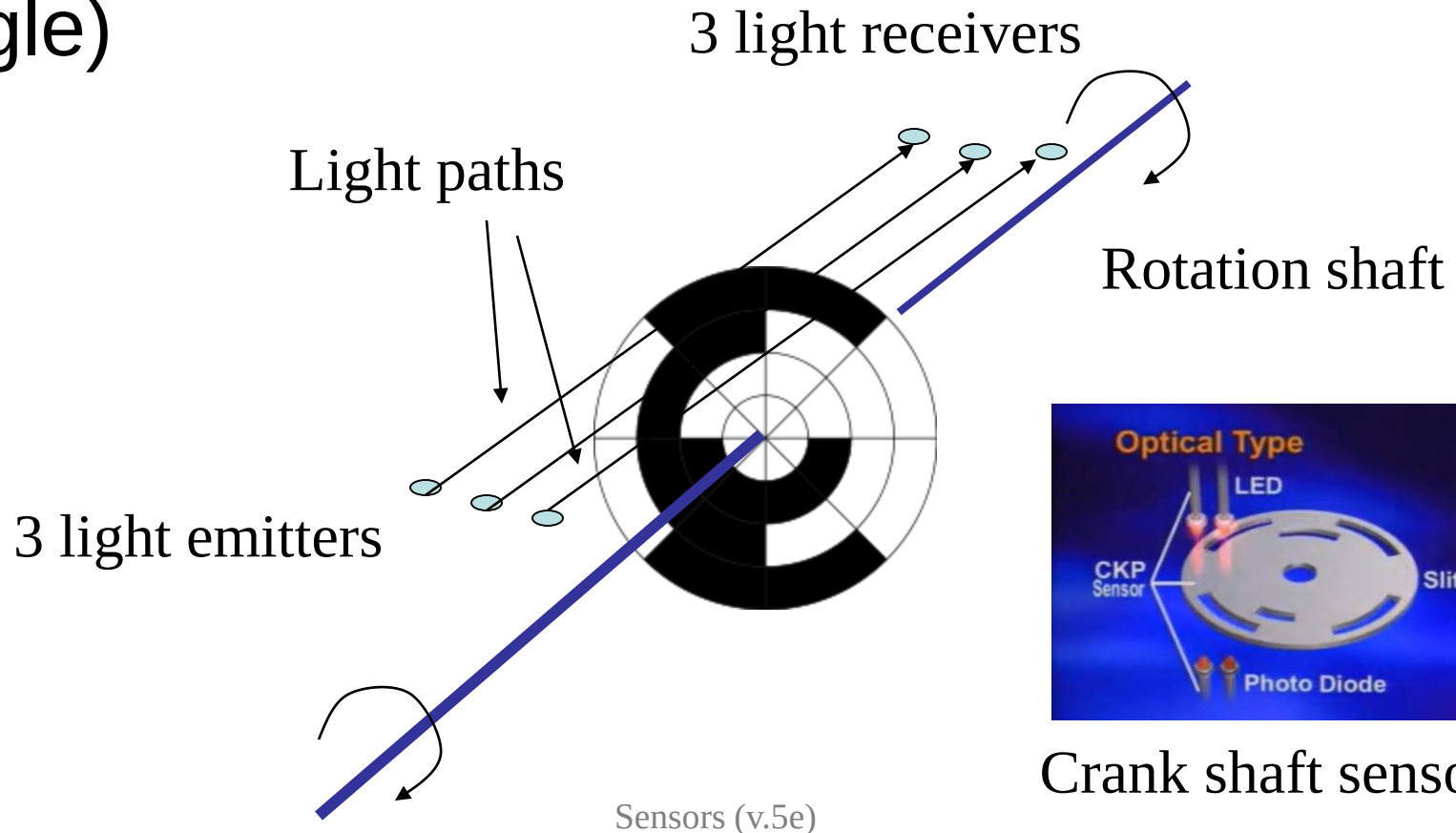
- non touch sensing



Optical rotary encoder

(http://en.wikipedia.org/wiki/Rotary_encoder)

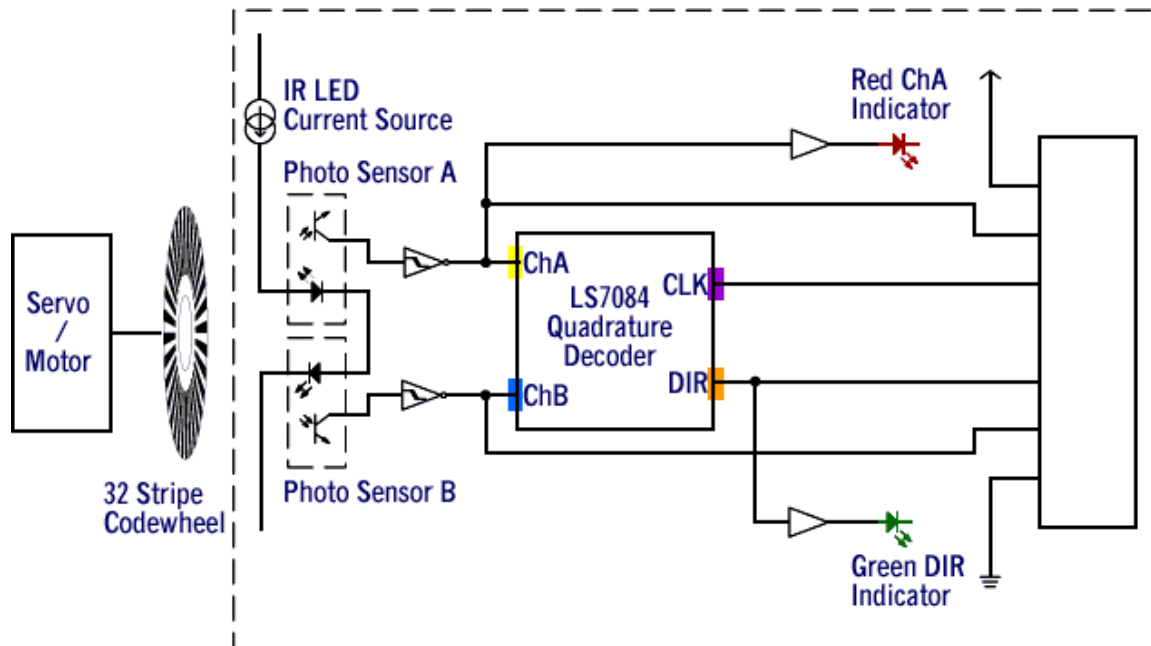
- The light received (on or off) will tell the rotation angle)



<http://www.youtube.com/watch?v=RuIisITGOwA>

Wheel Encoders

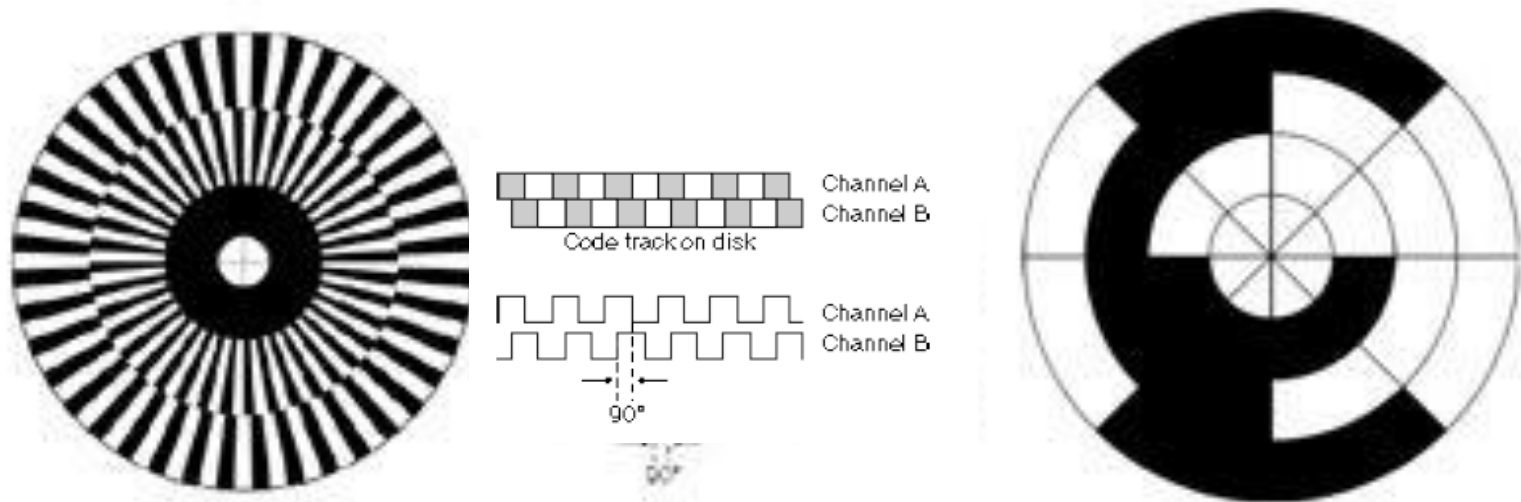
- Nubotics.com, \$27
- Jun 98, Oct 2000 Encoder



Wheel/Joint encoder

Optical encoder

To sense angular speed and position

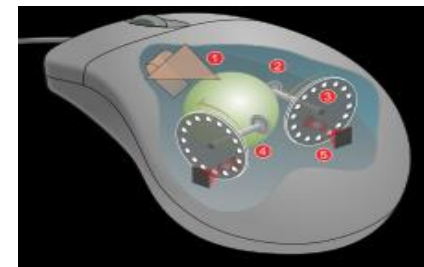
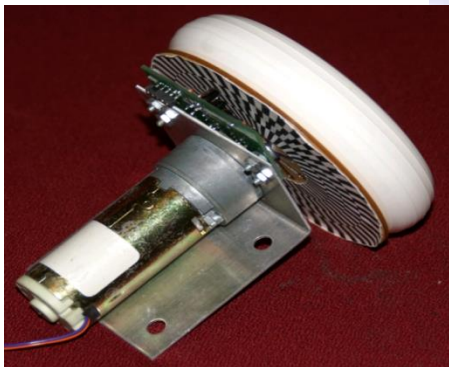


Quadrature encoder : pattern rotating with motor, and optical sensor that registers black/white transitions (to obtain direction)

Can also be implemented magnetically or electrically (same principle). Main stream technology: CNC machines and RC servos.

Wheel / Motor Encoders

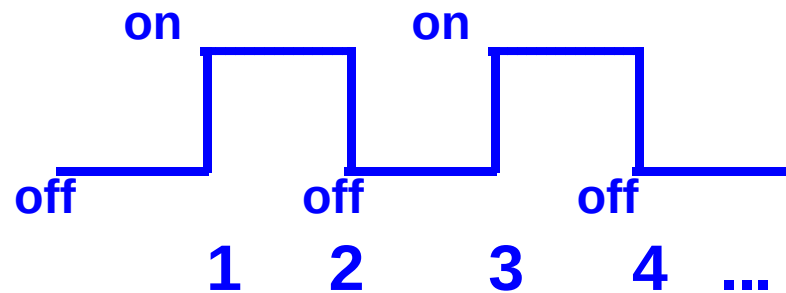
- measure position or speed of the wheels or steering
- integrate wheel movements to get an estimate of the position -> odometry
- optical encoders are proprioceptive sensors
- typical resolutions: 64 - 2048 increments per revolution.
 - for high resolution: interpolation



Encoders



Voltage square wave



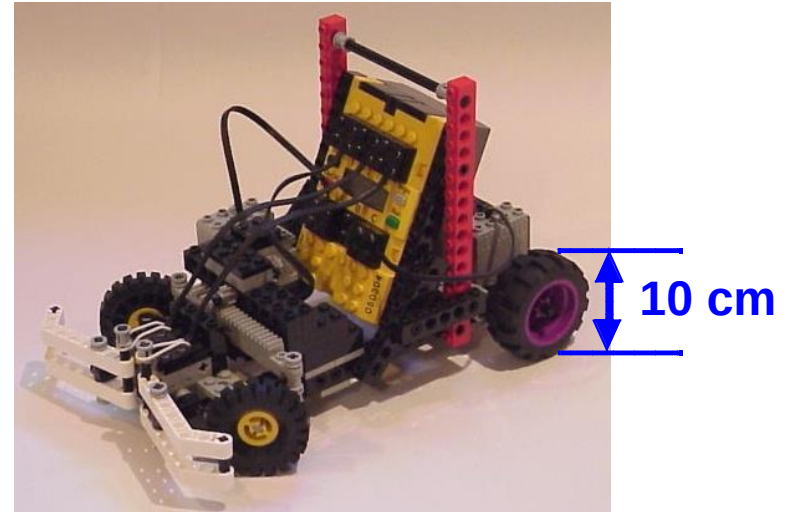
Important spec:
Number of counts
per revolution

Sample problems

Sensor Analysis



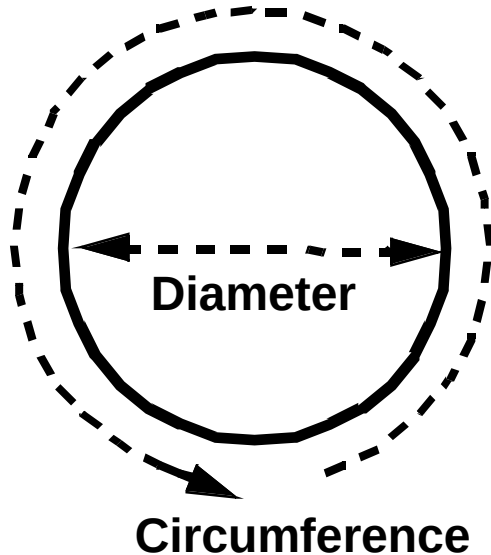
16 counts per rev.



10 cm wheel diameter

- ☐ How far does the wheel travel for 1 encoder count?
- ☐ What happens if we change the wheel diameter?
- ☐ How many counts are there per meter of travel?

Sample problems



Temos que numa volta temos 16 tics

$$C = \pi D$$

$$C = 10\pi \text{ cm}$$

isto é o quanto ele anda por cada tic

$$\frac{10\pi \text{ cm}}{1 \text{ rev}} \times \frac{1 \text{ rev}}{16 \text{ counts}} = \frac{1.96 \text{ cm}}{\text{count}}$$

vamos ter este
valor por cada
tic

Sample problems

Suppose I want 1.0 cm / count.

What should my wheel diameter be?

$$\frac{1.0 \text{ cm}}{\text{count}} \times \frac{16 \text{ counts}}{1 \text{ rev}} = \frac{16 \text{ cm}}{\text{rev}}$$

$$C = 16 \text{ cm}$$

$$D = \frac{C}{\pi} = \frac{16}{\pi} = 5.09 \text{ cm}$$

Sample problems

For my 10 cm wheel, how many encoder counts will there be for 1 meter of travel?

$$\frac{1.96 \text{ cm}}{\text{count}} \times \frac{1 \text{ meter}}{100 \text{ cm}} = \frac{0.0196 \text{ m}}{\text{count}}$$

$$\frac{1}{0.0196 \text{ m/ct}} = 51 \text{ counts/m}$$

(Active) Range Sensing strategies

- Active range sensors
- Ultrasound
- Laser range sensor

Sensors Based on EM Spectrum

- Basically used for ranging dá-nos os valores tipo analógicos (e não on and off)
- Light sensitive
 - eyes, cameras, photocells etc.
- Operating principle
 - CCD - charge coupled devices
 - photoelectric effect
- IR sensitive - FLIR sensíveis à luz infravermelha
 - sense heat differences and construct images
 - night vision application

EM Spectrum

- Radio and Microwave
 - RADAR: **R**adio **D**etection and **R**anging
 - Microwave radar: insensitive to clouds
- Coherent light
 - all photons have same phase and wavelength
 - LASER: **L**ight **A**mplification by **S**timulated **E**mission of **R**adiation
 - LASER RADAR: LADAR - accurate ranging

The SICK Laser Rangefinder



EM Spectrum

- Nuclear Magnetic Resonance (NMR)
 - heavy duty magnetic field lines up atoms in a body
 - now expose body to radio signals
 - different nuclei resonate at different frequencies which can be measured leading to an image

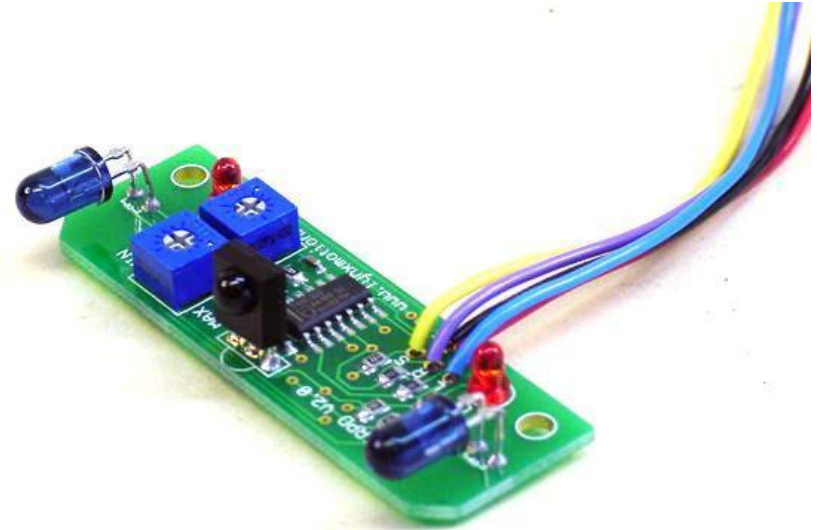
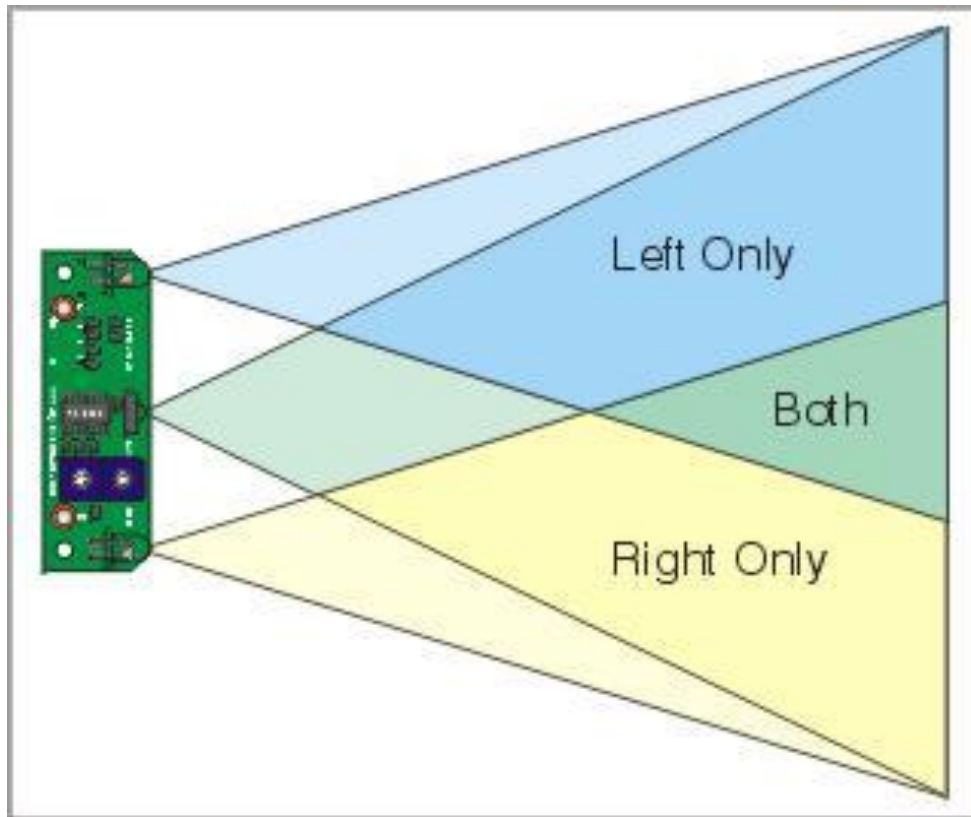
Local Proximity Sensing in EM

- Infrared LEDs
 - cheap, active sensing
 - usually low resolution - normally used for presence/absence of obstacles rather than ranging
 - operate over small range

Sensors – IR

- Active (emitting)
 - Oscillator generates IR reflections off objects
 - Filtered receiver looks for “reflections”
 - Pulses may be encoded for better discrimination
 - Typically frequencies around 40KHz
 - Doesn’t work well with dark, flat colored objects
- Passive (sensor only)
 - Pyro-electric (heat sensor)
 - Look for IR emissions from people & animals
 - Used in security systems & motion detectors

Infrared - Active



\$30 from Lynxmotion

July 99 Encoder

Active IR Sensor Specs

- Sensor type = Reflective IR
- IR detector = Panasonic PNA4602M
- IR LED type = Narrow focus 10°
- I/O required = 3 digital lines: 2 outputs, 1 input
- Range = Approximately 4" to 26"
- Input voltage = 5vdc regulated @ 8mA
- PC board size = 2.3" x .75"

Linear Array IR Range Sensors

- Sharp GP2Dxx (one of many)
- ~4 to 30cm Range
- Fixed Range with Discrete Output
- Analog or Digital Output
- Easy to Use



vai enviar uma onda e vai voltar a recebê-la. Através do tempo que existe entre enviar e receber, vai calcular a distância entre o sensor e o objeto -> Ao tempo chamamos de "tempo de voo".

Range sensors

Time of flight active range sensing

- Sonar

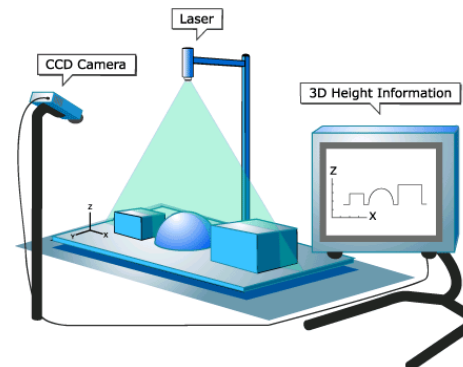


- Laser range finder
- Time of Flight Camera



Geometric active range sensor

- Structured light



Range Sensors (time of flight) (1)

- Range distance measurement: called range sensors
- Range information:
 - key element for localization and environment modeling
- Ultrasonic sensors as well as laser range sensors make use of propagation speed of sound or electromagnetic waves respectively. The traveled distance of a sound or electromagnetic wave is given by

distância que andei

$$d = c \cdot t$$

este é o tempo que a onda demora a voltar
velocidade da onda no meio

- Where
 - d = distance traveled (usually round-trip)
 - c = speed of wave propagation
 - t = time of flight.

Range Sensors (time of flight) (2)

- It is important to point out
 - Propagation speed v of sound: 0.3 m/ms
 - Propagation speed v of electromagnetic signals: 0.3 m/ns,
 - one million times faster.
 - 3 meters
 - is 10 ms ultrasonic system
 - only 10 ns for a laser range sensor
 - laser range sensors expensive and delicate
- The quality of time of flight range sensors mainly depends on:
 - Uncertainties about the exact time of arrival of the reflected signal
 - Inaccuracies in the time of flight measure (laser range sensors)
 - Opening angle of transmitted beam (ultrasonic range sensors)
 - Interaction with the target (surface, specular reflections)
 - Variation of propagation speed
 - Speed of mobile robot and target (if not at stand still)

Sensors Based on Sound

- SONAR: **S**ound **N**avigation **a**nd **R**anging
 - bounce sound off of something
 - measure time for reflection to be heard - gives a range measurement
 - measure change in frequency - gives the relative speed of the object (Doppler effect)
 - bats and dolphins use it with amazing results
 - robots use it w/ less than amazing results

Sensors – Ultrasonic

- Active
 - Emit pulses & listen for echos
 - Times round trip sound travel ($\sim 1\text{ft/mS}$)
 - Reaches far fairly beyond robot (inches to 30-50')
 - Relatively simple, not cheap, analog output
 - Directional; not everything reflects sound well
- Passive (listens only)
 - Sensor listens for ultrasonic sounds
 - Electronics may translate frequency or modulation
 - Software may perform signal analysis (FFTs, etc.)

Ultrasonic - Active



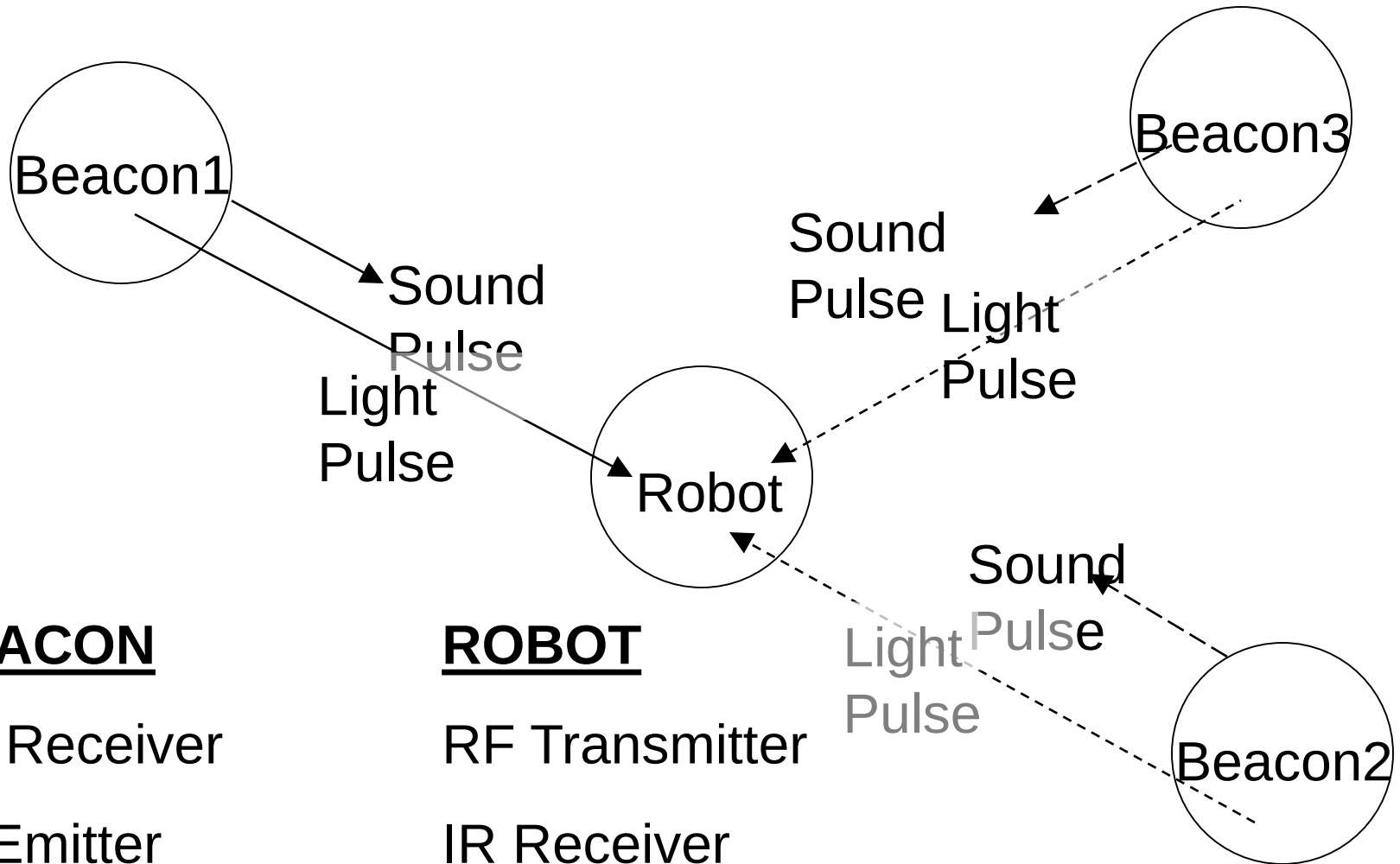
Visit <http://www.acroname.com> for more information about these & other products. Search the web for “polaroid ultrasonic sensor”

Sensors – Ultrasonic (cont.)

- Passive - Beacons & Sensors
 - Beacons listen: RF command to broadcast
 - Send light & sound pulses
 - Robot looks & listens for each beacon
 - Light pulse starts timer, sound pulse stops it
 - Robot knows location of each beacon
 - Compass on robot provides its orientation
 - Robot computes distance, measures bearing
 - Robot can then compute its location

(Speed of Light=1 ft/nS, Speed of Sound=1ft/mS)

Ultrasonic - Passive



BEACON

RF Receiver

IR Emitter

Ultrasonic Emitter

ROBOT

RF Transmitter

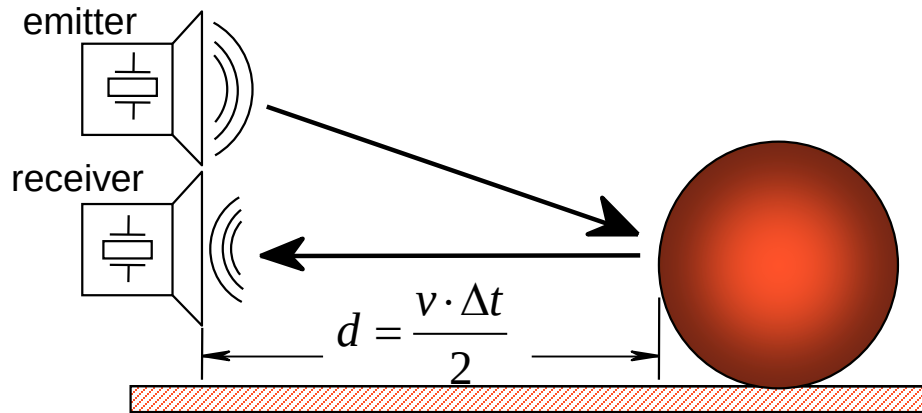
IR Receiver

Ultrasonic Receiver

Sensors – Sonic (Acoustic)

- Active
 - Emit pulses & listen for echos
 - Times round trip sound travel ($\sim 1\text{ft/mS}$)
 - Reaches far fairly beyond robot (30-50 ft)
 - Relatively simple, not cheap, analog output
 - Directional, not everything reflects sound
 - Noisy!!!!
- Passive (sensor only)
 - Sensor listens to ambient sounds
 - Filters or scans selected frequencies
 - ADC measures conditioned signal amplitude
 - CPU performs signal analysis on what it hears

Factsheet: Ultrasonic Range Sensor



<http://www.robot-electronics.co.uk/shop/Ultrasonic_Rangers1999.htm>

1. Operational Principle

An ultrasonic pulse is generated by a piezo-electric emitter, reflected by an object in its path, and sensed by a piezo-electric receiver. Based on the speed of sound in air and the elapsed time from emission to reception, the distance between the sensor and the object is easily calculated.

2. Main Characteristics

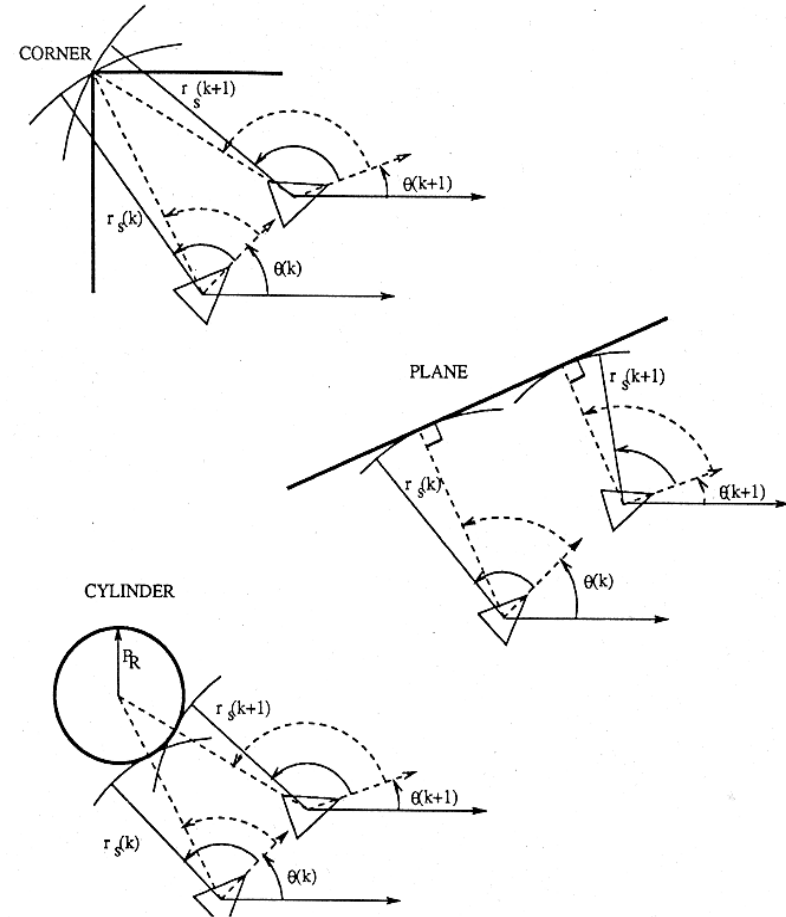
- Precision influenced by angle to object (as illustrated on the next slide)
- Useful in ranges from several cm to several meters
- Typically relatively inexpensive

3. Applications

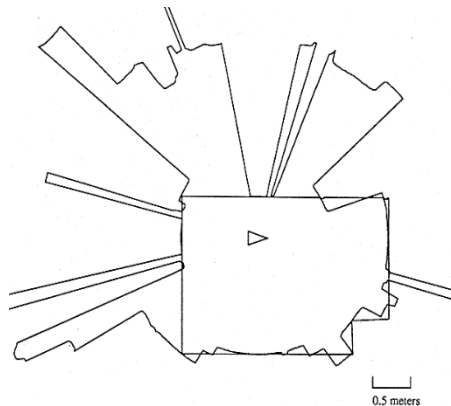
- Distance measurement (also for transparent surfaces)
- Collision detection

Ultrasonic Sensor (sound)

- Other problems for ultrasonic sensors
 - soft surfaces that **absorb** most of the sound energy
 - surfaces that are far from being perpendicular to the direction of the sound -> **specular reflection**



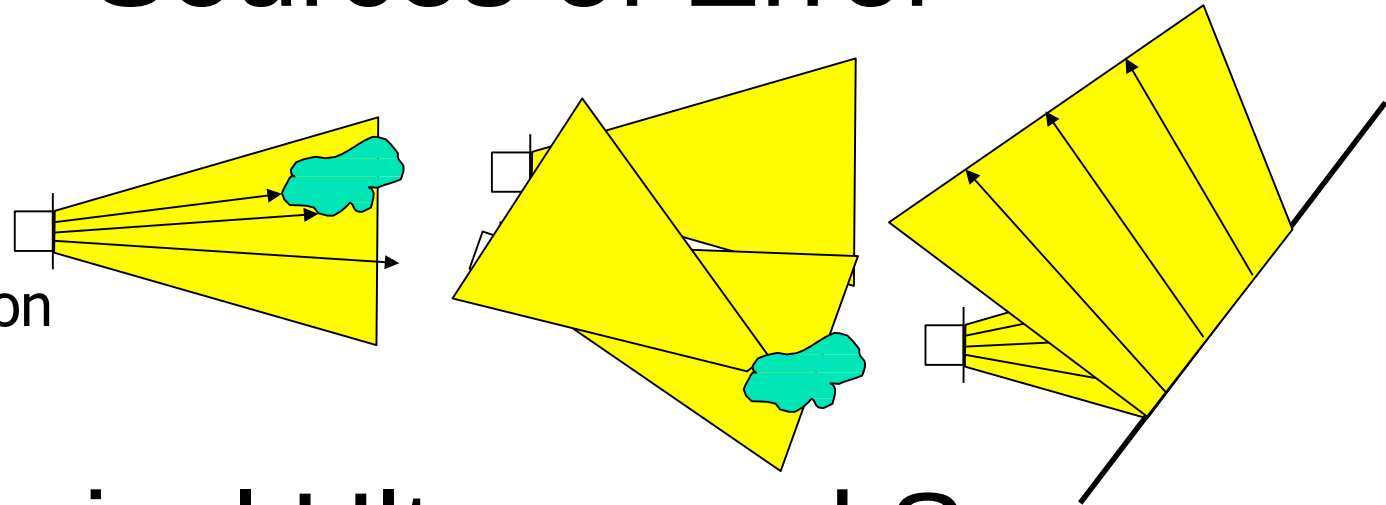
a) 360° scan



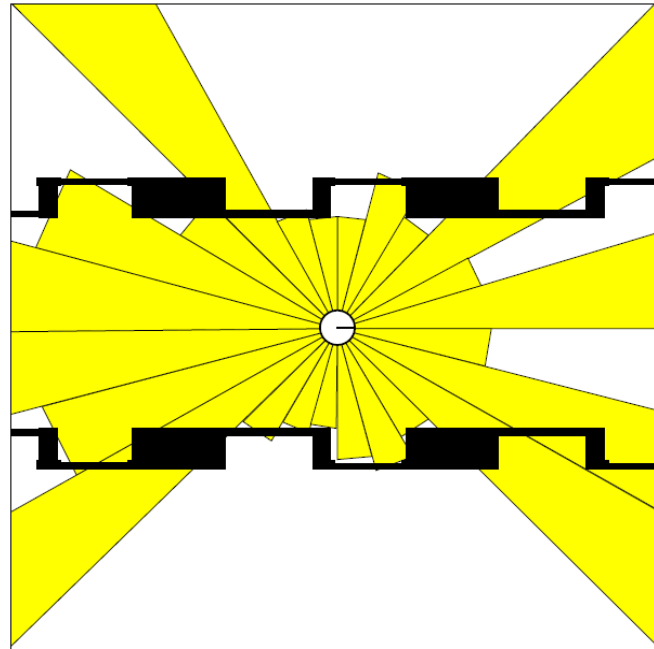
b) results from different geometric primitives

Sources of Error

- Opening angle
- Crosstalk
- Specular reflection



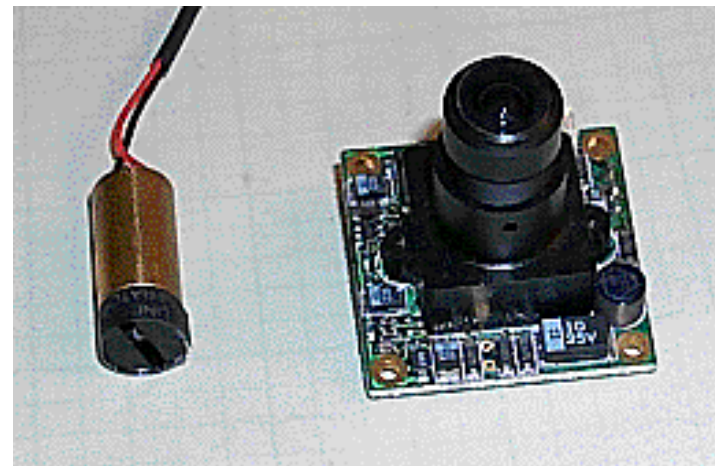
Typical Ultrasound Scan



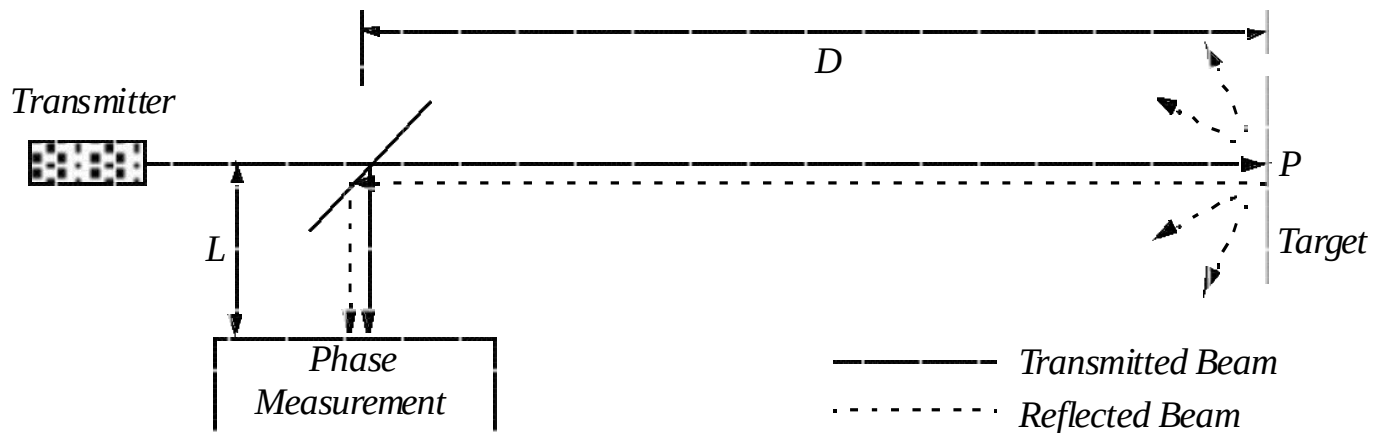
Slide adopted from C. Stachniss

Laser Range Sensors

- USB Interface
- 240° Field of View
- 0.36° Angular Resolution
- 10Hz Refresh Rate
- 20mm to 4m
- \$2695 (cool but pricey)
- Also See:
 - Oct 2001 Encoder
 - Kenneth Maxon

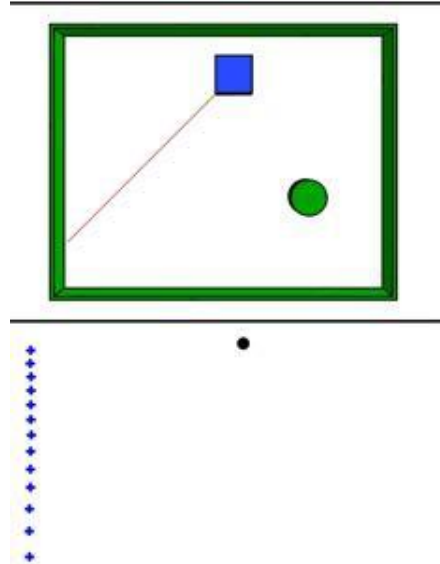
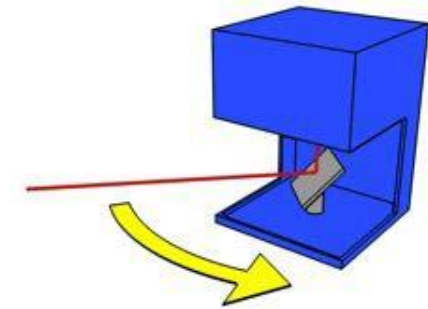
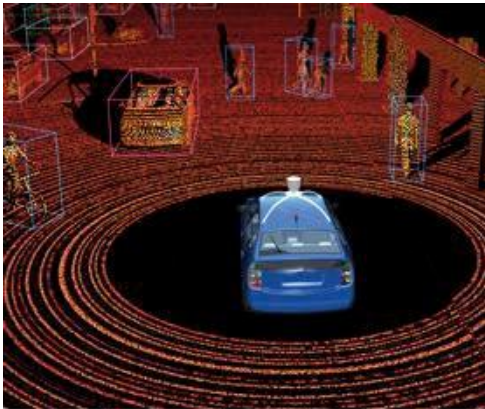


Laser Range Sensor (electromagnetic)

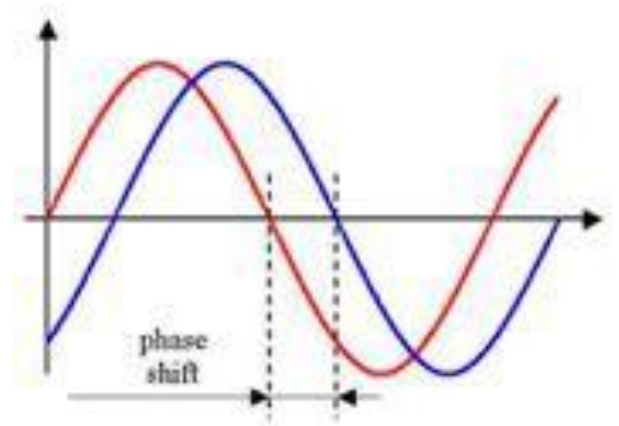
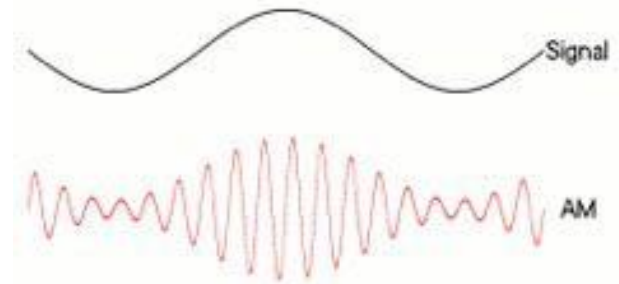


- Laser light instead of sound
- Transmitted and received beams coaxial
- Transmitter illuminates a target with a collimated beam
- Receiver detects the time needed for round-trip
- Lidar (light detection and ranging)

Distance from phase shift



5 Mhz ~ 60m



Laser Range Finder: Applications



Stanley: Stanford
(winner of the 2005 Darpa Grand Challenge)
5 Sick laser scanners

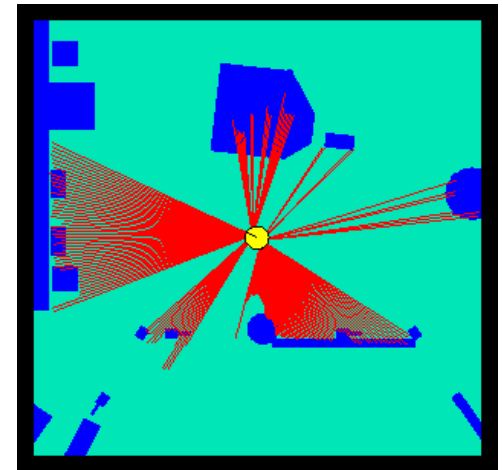
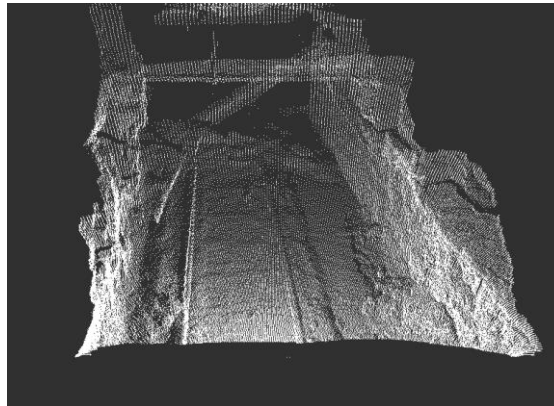
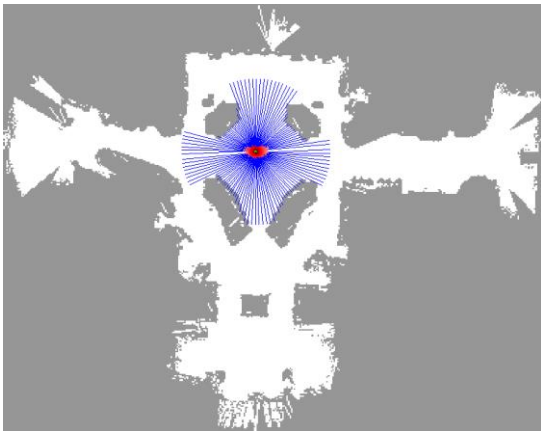
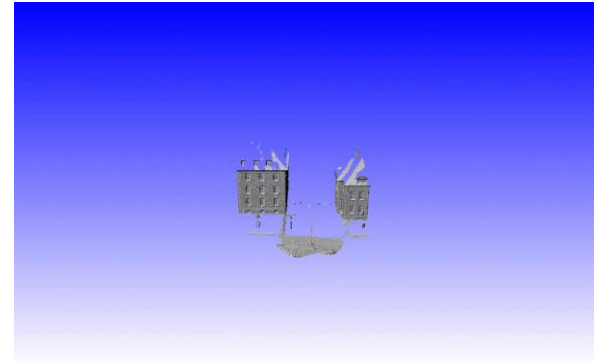
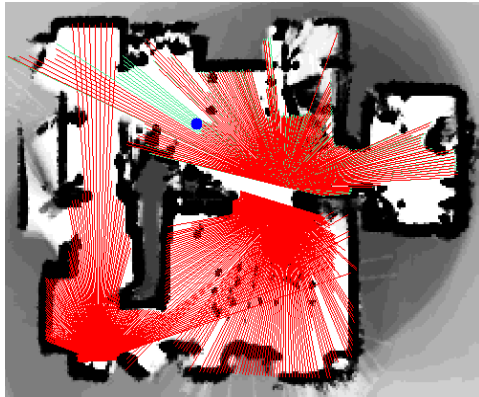
Autonomous Smart:
ASL ETH Zurich



Robots Equipped with Laser Scanners



Typical Scans

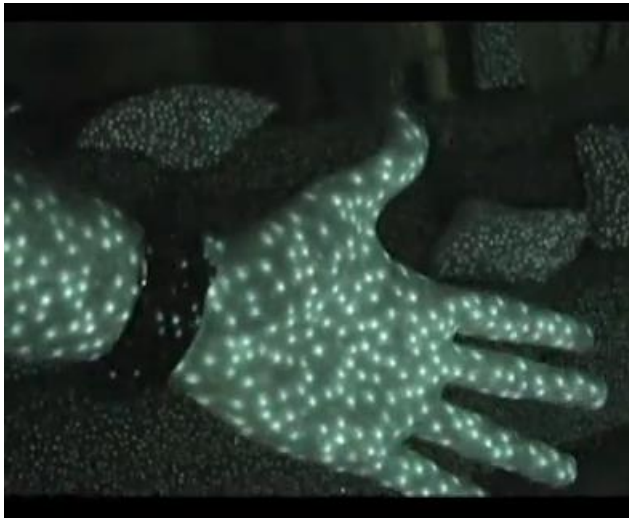


Novel

- Kinect



<http://www.ladyada.net/learn/diykinect/>

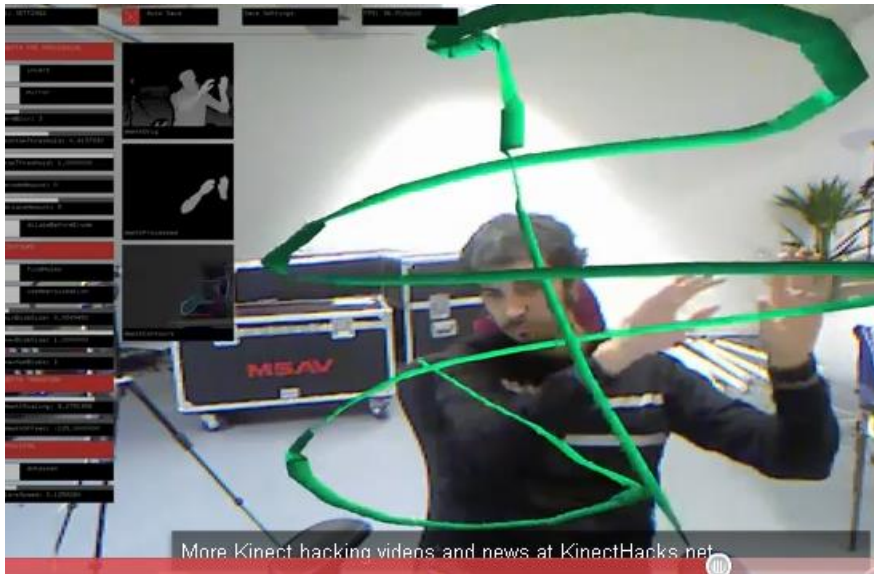


<http://www.youtube.com/watch?v=nvvQJxgykcU>



<http://www.youtube.com/watch?v=Mf44bWQr3jc>

Many KINECT DIY projects



<http://www.youtube.com/watch?v=Brpu30vjCa4&feature=related>

<http://www.youtube.com/watch?v=3V-TGXFZbbA&feature=related>

Problem: Sensor Choice

- What sensors to employ ?
- E.g. mapping
 - ranging - laser, sonar, IR, stereo camera pair
 - salient feature detection - doors using color
- Factors
 - accuracy, cost, information needed etc etc.

Problem: Sensor Placement

- Where to put them ?
- On/off board (e.g. localization using odometry vs. localization using beacons)
- If onboard - where ?
 - Reasonable arrangements - heuristic
 - Optimal arrangements - mathematically rigorous

Characterizing Sensor Performance

- Basic sensor response ratings (cont.)

- Resolution

- minimum difference between two values
 - usually: lower limit of dynamic range = resolution
 - for digital sensors it is usually the A/D resolution.
 - e.g. 5V / 255 (8 bit)

- Linearity

- variation of output signal as function of the input signal
 - linearity is less important when signal is after treated with a computer

- Bandwidth or Frequency

- the speed with which a sensor can provide a stream of readings
 - usually there is an upper limit depending on the sensor and the sampling rate
 - Lower limit is also possible, e.g. acceleration sensor

In Situ Sensor Performance (1)

Characteristics that are relevant for real world environments

- Sensitivity

- ratio of output change to input change
- however, in real world environment, the sensor has very often high sensitivity to other environmental changes, e.g. illumination

- Cross-sensitivity

- sensitivity to environmental parameters that are orthogonal to the target parameters

- Error / Accuracy

- difference between the sensor's output and the true value

$$\left(accuracy = 1 - \frac{|m - v|}{v} \right) \quad \text{error} \quad \begin{array}{l} \mathbf{m = measured} \\ \mathbf{v = true value} \end{array}$$

In Situ Sensor Performance (2)

- Characteristics that are especially relevant for real world environments
- Systematic error -> deterministic errors
 - caused by factors that can (in theory) be modeled -> prediction
 - e.g. calibration of a laser sensor or of the distortion caused by the optic of a camera
- Random error -> non-deterministic
 - no prediction possible
 - however, they can be described probabilistically
 - e.g. Hue instability of camera, black level noise of camera ..
- Precision
 - reproducibility of sensor results

Characterizing Error: The Challenges in Mobile Robotics

- Robots have to perceive, analyze and interpret the state of the surrounding
- Measurements in real world environment are dynamically changing and error prone.
- Examples:
 - changing illuminations
 - specular reflections
 - light or sound absorbing surfaces
 - cross-sensitivity of robot sensor to robot pose and robot- environment dynamics
 - rarely possible to model -> appear as random errors
 - systematic errors and random errors might be well defined in controlled environment.
This is not the case for robots !!

Multi-Modal Error Distributions: The Challenges in ...

- Behavior of sensors modeled by probability distribution (random errors)
 - usually very little knowledge about the causes of random errors
 - often probability distribution is assumed to be symmetric or even Gaussian
 - however, it is important to realize how wrong this can be!
 - Examples:
 - Sonar (ultrasonic) sensor might overestimate the distance in real environment and is therefore not symmetric
 - Thus the sonar sensor might be best modeled by two modes:
 - mode for the case that the signal returns directly
 - mode for the case that the signals returns after multi- path reflections.
 - Stereo vision system might correlate to images incorrectly, thus causing results that make no sense at all

Autonomous Robots

- The control of autonomous robots involves a number of subtasks
 - Understanding and modeling of the mechanism
 - Kinematics, Dynamics, and Odometry
 - Reliable control of the actuators
 - Closed-loop control
 - Generation of task-specific motions
 - Path planning
 - Integration of sensors
 - Selection and interfacing of various types of sensors
 - Coping with noise and uncertainty
 - Filtering of sensor noise and actuator uncertainty
 - Creation of flexible control policies
 - Control has to deal with new situations

Requirements for Robots in Intelligent Environments

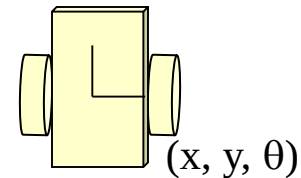
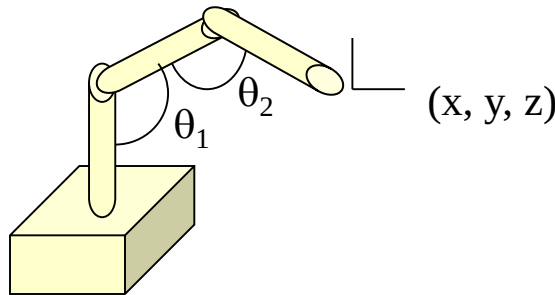
- Autonomy
 - Robots have to be capable of achieving task objectives without human input
 - Robots have to be able to make and execute their own decisions based on sensor information
- Intuitive Human-Robot Interfaces
 - Use of robots in smart homes can not require extensive user training
 - Commands to robots should be natural for inhabitants
- Adaptation
 - Robots have to be able to adjust to changes in the environment

Autonomous Robot Control

- To control robots to perform tasks autonomously a number of tasks have to be addressed:
 - Modeling of robot mechanisms
 - Kinematics, Dynamics
 - Robot sensor selection
 - Active and passive proximity sensors
 - Low-level control of actuators
 - Closed-loop control
 - Control architectures
 - Traditional planning architectures
 - Behavior-based control architectures
 - Hybrid architectures

Modeling the Robot Mechanism

- Forward kinematics describes how the robots joint angle configurations translate to locations in the world



- Inverse kinematics computes the joint angle configuration necessary to reach a particular point in space.
- Jacobians calculate how the speed and configuration of the actuators translate into velocity of the robot

Mobile Robot Odometry

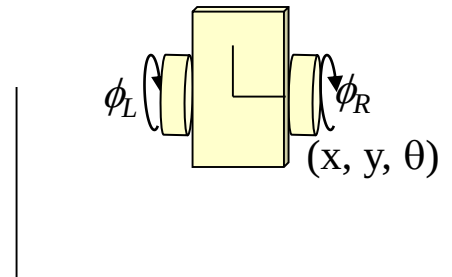
- In mobile robots the same configuration in terms of joint angles does not identify a unique location
 - To keep track of the robot it is necessary to incrementally update the location (this process is called odometry or dead reckoning)

$$\begin{pmatrix} x \\ y \\ \theta \end{pmatrix}^{t+\Delta t} = \begin{pmatrix} x \\ y \\ \theta \end{pmatrix}^t + \begin{pmatrix} v_x \\ v_y \\ \varpi \end{pmatrix} \Delta t$$

- Example: A differential drive robot

$$v_x = \cos(\theta) \frac{r(\dot{\phi}_L + \dot{\phi}_R)}{2}, v_y = \sin(\theta) \frac{r(\dot{\phi}_L + \dot{\phi}_R)}{2}$$

$$\varpi = \frac{r}{d} (\dot{\phi}_L - \dot{\phi}_R)$$



Actuator Control

- To get a particular robot actuator to a particular location it is important to apply the correct amount of force or torque to it.
 - Requires knowledge of the dynamics of the robot
 - Mass, inertia, friction
 - For a simplistic mobile robot: $F = m a + B v$
 - Frequently actuators are treated as if they were independent (i.e. as if moving one joint would not affect any of the other joints).
 - The most common control approach is PD-control (proportional, differential control)
 - For the simplistic mobile robot moving in the x direction:

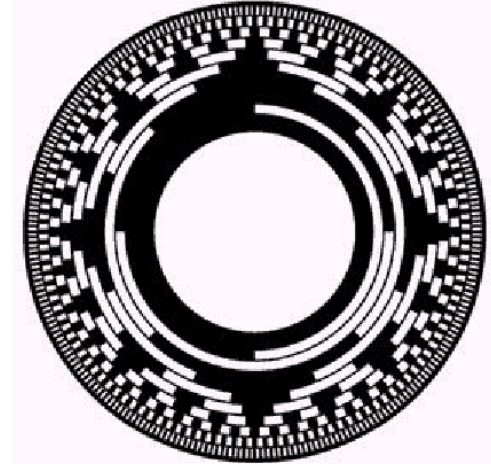
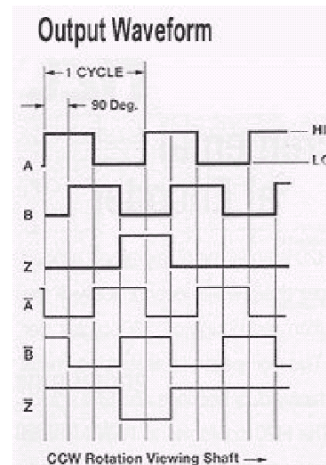
$$F = K_P (x_{desired} - x_{actual}) + K_D (v_{desired} - v_{actual})$$

Sensor-Driven Robot Control

- To accurately achieve a task in an intelligent environment, a robot has to be able to react dynamically to changes in its surrounding
 - Robots need sensors to perceive the environment
 - Most robots use a set of different sensors
 - Different sensors serve different purposes
 - Information from sensors has to be integrated into the control of the robot

Robot Sensors

- Internal sensors to measure the robot configuration
 - Encoders measure the rotation angle of a joint



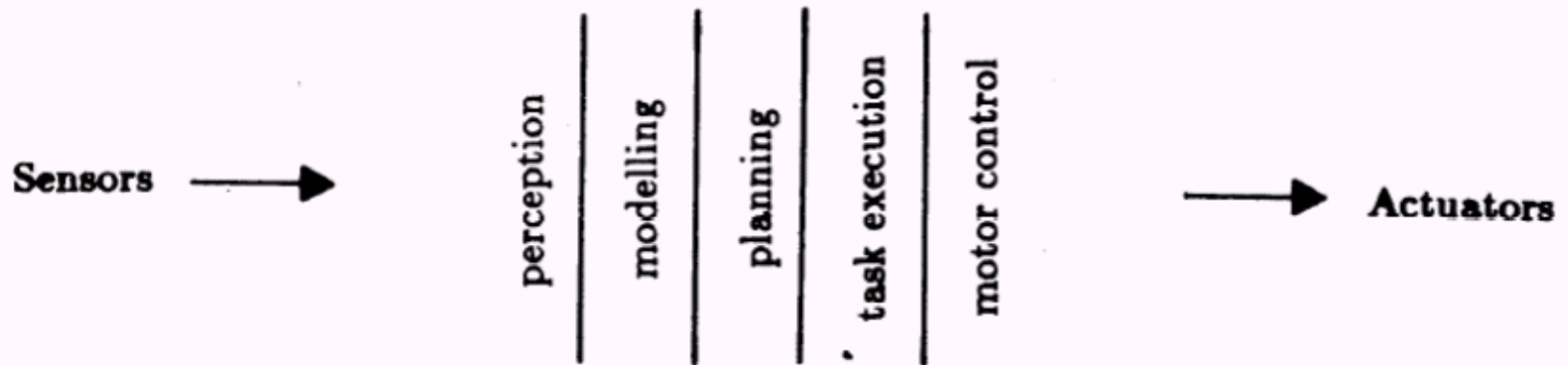
- Limit switches detect when the joint has reached the limit

Uncertainty in Robot Systems

- Robot systems in intelligent environments have to deal with sensor noise and uncertainty
 - Sensor uncertainty
 - Sensor readings are imprecise and unreliable
 - Non-observability
 - Various aspects of the environment can not be observed
 - The environment is initially unknown
 - Action uncertainty
 - Actions can fail
 - Actions have nondeterministic outcomes

Deliberative Robot Control Architectures

- In a deliberative control architecture the robot first plans a solution for the task by reasoning about the outcome of its actions and then executes it



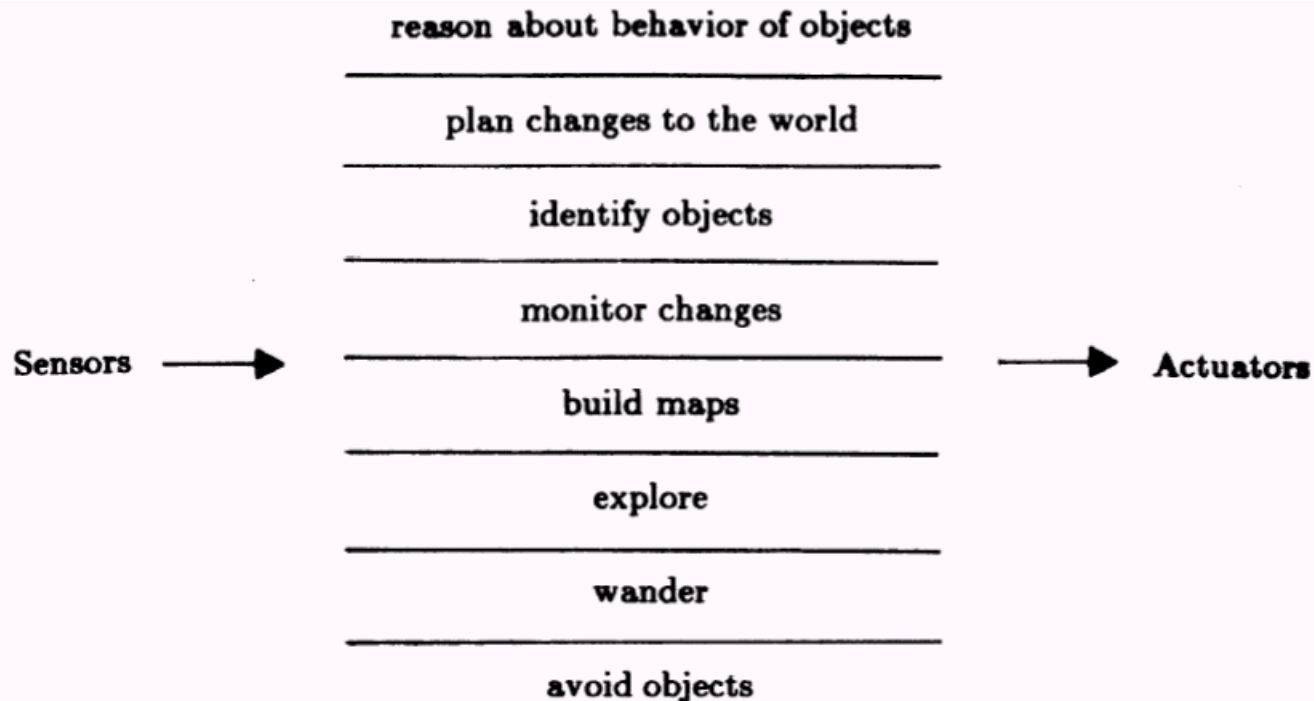
- Control process goes through a sequence of sensing, model update, and planning steps

Deliberative Control Architectures

- Advantages
 - Reasons about contingencies
 - Computes solutions to the given task
 - Goal-directed strategies
- Problems
 - Solutions tend to be fragile in the presence of uncertainty
 - Requires frequent replanning
 - Reacts relatively slowly to changes and unexpected occurrences

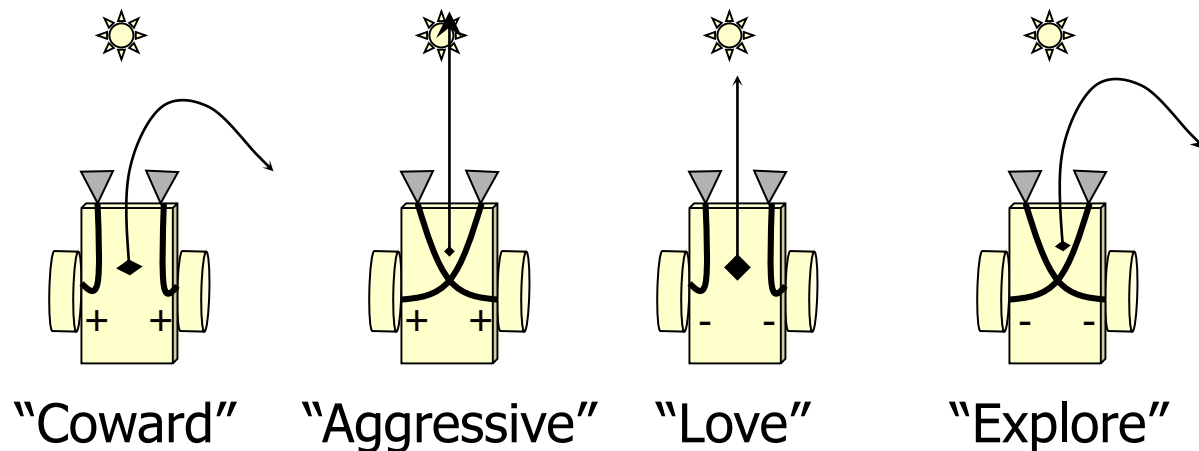
Behavior-Based Robot Control Architectures

- In a behavior-based control architecture the robot's actions are determined by a set of parallel, reactive behaviors which map sensory input and state to actions.



Complex Behavior from Simple Elements: Braitenberg Vehicles

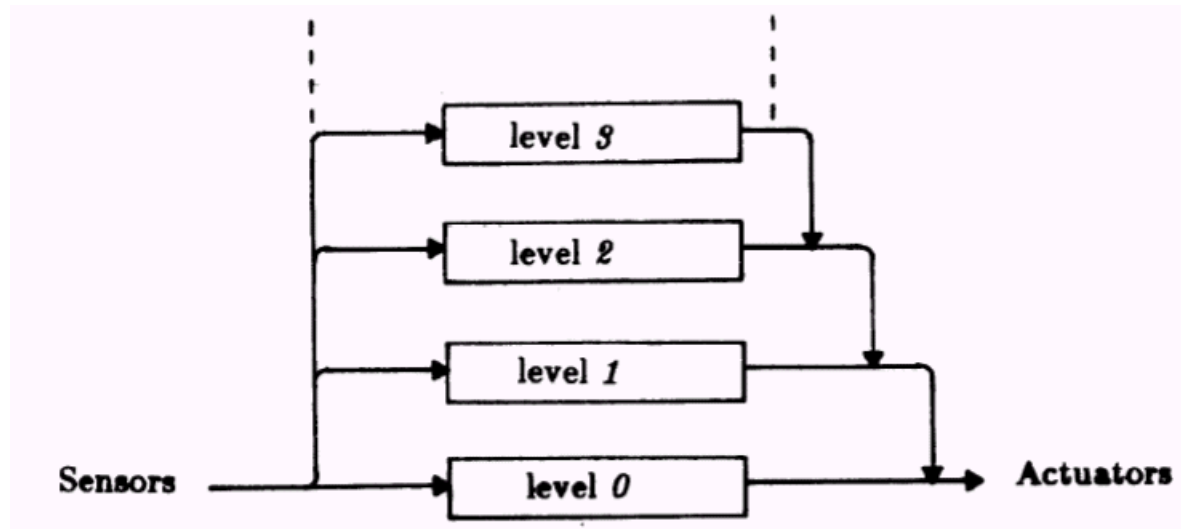
- Complex behavior can be achieved using very simple control mechanisms
 - Braitenberg vehicles: differential drive mobile robots with two light sensors



- Complex external behavior does not necessarily require a complex reasoning mechanism

Behavior-Based Architectures: Subsumption Example

- Subsumption architecture is one of the earliest behavior-based architectures
 - Behaviors are arranged in a strict priority order where higher priority behaviors subsume lower priority ones as long as they are not inhibited.

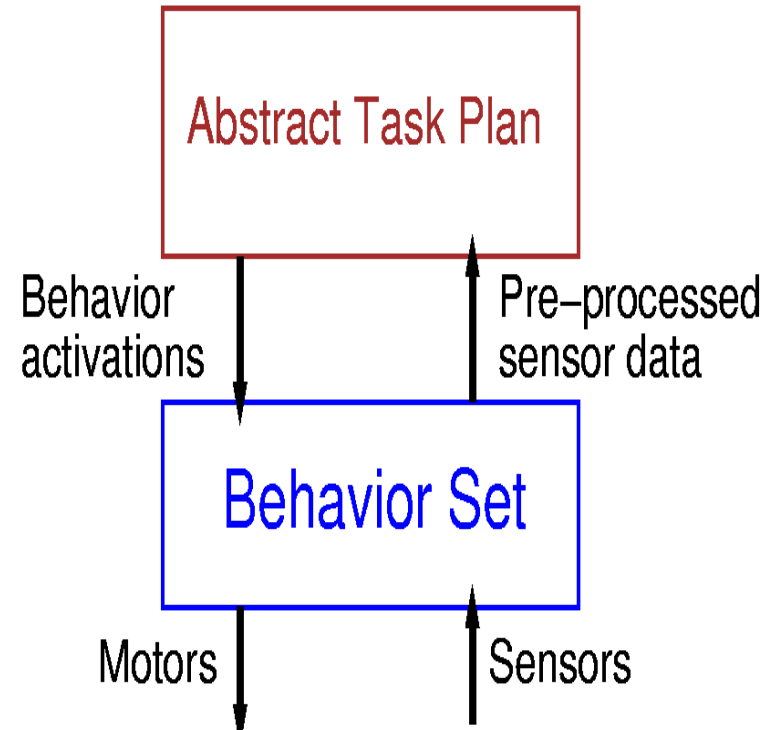


Reactive, Behavior-Based Control Architectures

- Advantages
 - Reacts fast to changes
 - Does not rely on accurate models
 - “The world is its own best model”
 - No need for replanning
- Problems
 - Difficult to anticipate what effect combinations of behaviors will have
 - Difficult to construct strategies that will achieve complex, novel tasks
 - Requires redesign of control system for new tasks

Hybrid Control Architectures

- Hybrid architectures combine reactive control with abstract task planning
 - Abstract task planning layer
 - Deliberative decisions
 - Plans goal directed policies
 - Reactive behavior layer
 - Provides reactive actions
 - Handles sensors and actuators



Hybrid Control Architectures

- Advantages
 - Permits goal-based strategies
 - Ensures fast reactions to unexpected changes
 - Reduces complexity of planning
- Problems
 - Choice of behaviors limits range of possible tasks
 - Behavior interactions have to be well modeled to be able to form plans

Traditional Human-Robot Interface: Teleoperation

- Remote Teleoperation: Direct operation of the robot by the user
 - User uses a 3-D joystick or an exoskeleton to drive the robot
 - Simple to install
 - Removes user from dangerous areas
 - Problems:
 - Requires insight into the mechanism
 - Can be exhaustive
 - Easily leads to operation errors



Intuitive Robot Interfaces: Command Input

- Graphical programming interfaces
 - Users construct policies from elemental blocks
 - Problems:
 - Requires substantial understanding of the robot
- Deictic (pointing) interfaces
 - Humans point at desired targets in the world or
 - Target specification on a computer screen
 - Problems:
 - How to interpret human gestures ?
- Voice recognition
 - Humans instruct the robot verbally
 - Problems:
 - Speech recognition is very difficult
 - Robot actions corresponding to words has to be defined

Intuitive Robot Interfaces: Robot-Human Interaction

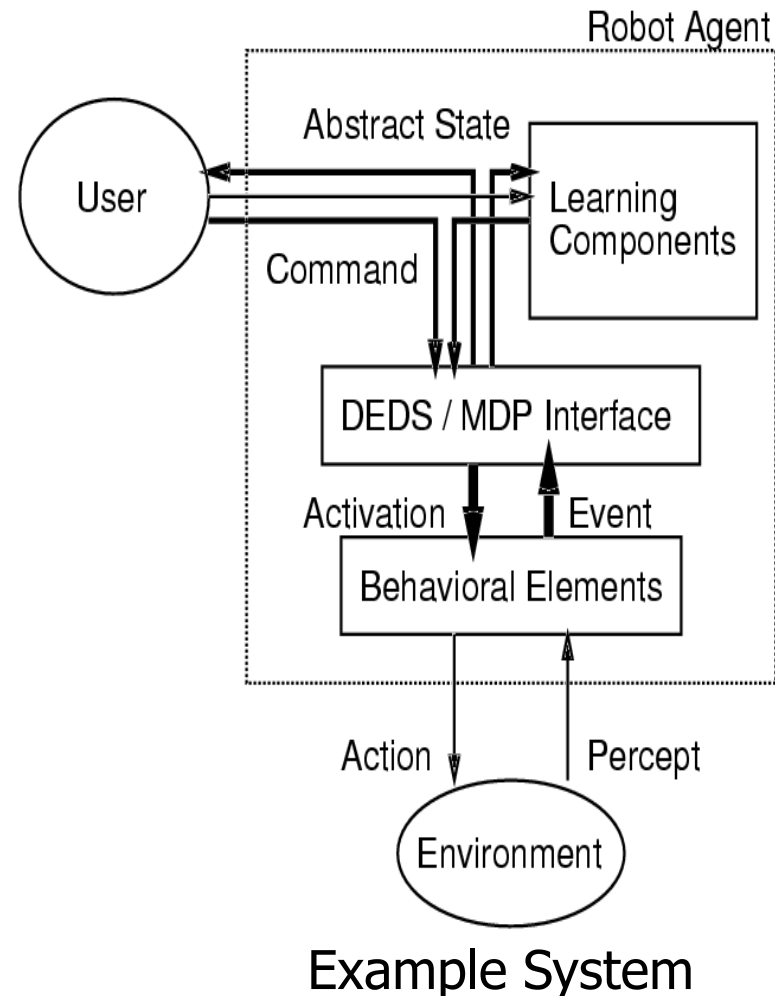
- The robot has to be able to communicate its intentions to the human
 - Output has to be easy to understand by humans
 - Robot has to be able to encode its intention
 - Interface has to keep human's attention without annoying her
- Robot communication devices:
 - Easy to understand computer screens
 - Speech synthesis
 - Robot “gestures”

Human-Robot Interfaces

- Existing technologies
 - Simple voice recognition and speech synthesis
 - Gesture recognition systems
 - On-screen, text-based interaction
- Research challenges
 - How to convey robot intentions ?
 - How to infer user intent from visual observation (how can a robot imitate a human) ?
 - How to keep the attention of a human on the robot ?
 - How to integrate human input with autonomous operation ?

Integration of Commands and Autonomous Operation

- Adjustable Autonomy
 - The robot can operate at varying levels of autonomy
 - Operational modes:
 - Autonomous operation
 - User operation / teleoperation
 - Behavioral programming
 - Following user instructions
 - Imitation
 - Types of user commands:
 - Continuous, low-level instructions (teleoperation)
 - Goal specifications
 - Task demonstrations



"Social" Robot Interactions

- To make robots acceptable to average users they should appear and behave “natural”
 - "Attentional" Robots
 - Robot focuses on the user or the task
 - Attention forms the first step to imitation
 - "Emotional" Robots
 - Robot exhibits “emotional” responses
 - Robot follows human social norms for behavior
 - Better acceptance by the user (users are more forgiving)
 - Human-machine interaction appears more “natural”
 - Robot can influence how the human reacts

Human-Robot Interfaces for Intelligent Environments

- Robot Interfaces have to be easy to use
 - Robots have to be controllable by untrained users
 - Robots have to be able to interact not only with their owner but also with other people
- Robot interfaces have to be usable at the human's discretion
 - Human-robot interaction occurs on an irregular basis
 - Frequently the robot has to operate autonomously
 - Whenever user input is provided the robot has to react to it
- Interfaces have to be designed human-centric
 - The role of the robot is it to make the human's life easier and more comfortable (it is not just a tech toy)

Adaptation and Learning for Robots in Smart Homes

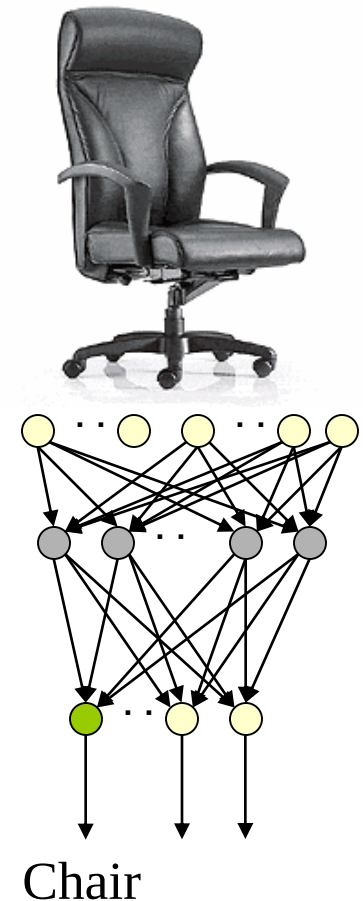
- Intelligent Environments are non-stationary and change frequently, requiring robots to adapt
 - Adaptation to changes in the environment
 - Learning to address changes in inhabitant preferences
- Robots in intelligent environments can frequently not be pre-programmed
 - The environment is unknown
 - The list of tasks that the robot should perform might not be known beforehand
 - No proliferation of robots in the home
 - Different users have different preferences

Learning Approaches for Robot Systems

- Supervised learning by teaching
 - Robots can learn from direct feedback from the user that indicates the correct strategy
 - The robot learns the exact strategy provided by the user
- Learning from demonstration (Imitation)
 - Robots learn by observing a human or a robot perform the required task
 - The robot has to be able to “understand” what it observes and map it onto its own capabilities
- Learning by exploration
 - Robots can learn autonomously by trying different actions and observing their results
 - The robot learns a strategy that optimizes reward

Learning Sensory Patterns

- Learning to Identify Objects
 - How can a particular object be recognized ?
 - Programming recognition strategies is difficult because we do not fully understand how we perform recognition
 - Learning techniques permit the robot system to form its own recognition strategy
 - Supervised learning can be used by giving the robot a set of pictures and the corresponding classification
 - Neural networks
 - Decision trees



Learning Task Strategies by Experimentation

- Autonomous robots have to be able to learn new tasks even without input from the user
 - Learning to perform a task in order to optimize the reward the robot obtains (Reinforcement Learning)
 - Reward has to be provided either by the user or the environment
 - Intermittent user feedback
 - Generic rewards indicating unsafe or inconvenient actions or occurrences
 - The robot has to explore its actions to determine what their effects are
 - Actions change the state of the environment
 - Actions achieve different amounts of reward
 - During learning the robot has to maintain a level of safety

Conclusions

- Robots are an important component in Intelligent Environments
 - Automate devices
 - Provide physical services
- Robot Systems in these environments need particular capabilities
 - Autonomous control systems
 - Simple and natural human-robot interface
 - Adaptive and learning capabilities
 - Robots have to maintain safety during operation
- While a number of techniques to address these requirements exist, no functional, satisfactory solutions have yet been developed
 - Only very simple robots for single tasks in intelligent environments exist

References

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